

Information Retrieval

3. Term vocabulary

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Spring 2022

Stock picture 123RF / ILYA AKINSHIN

3

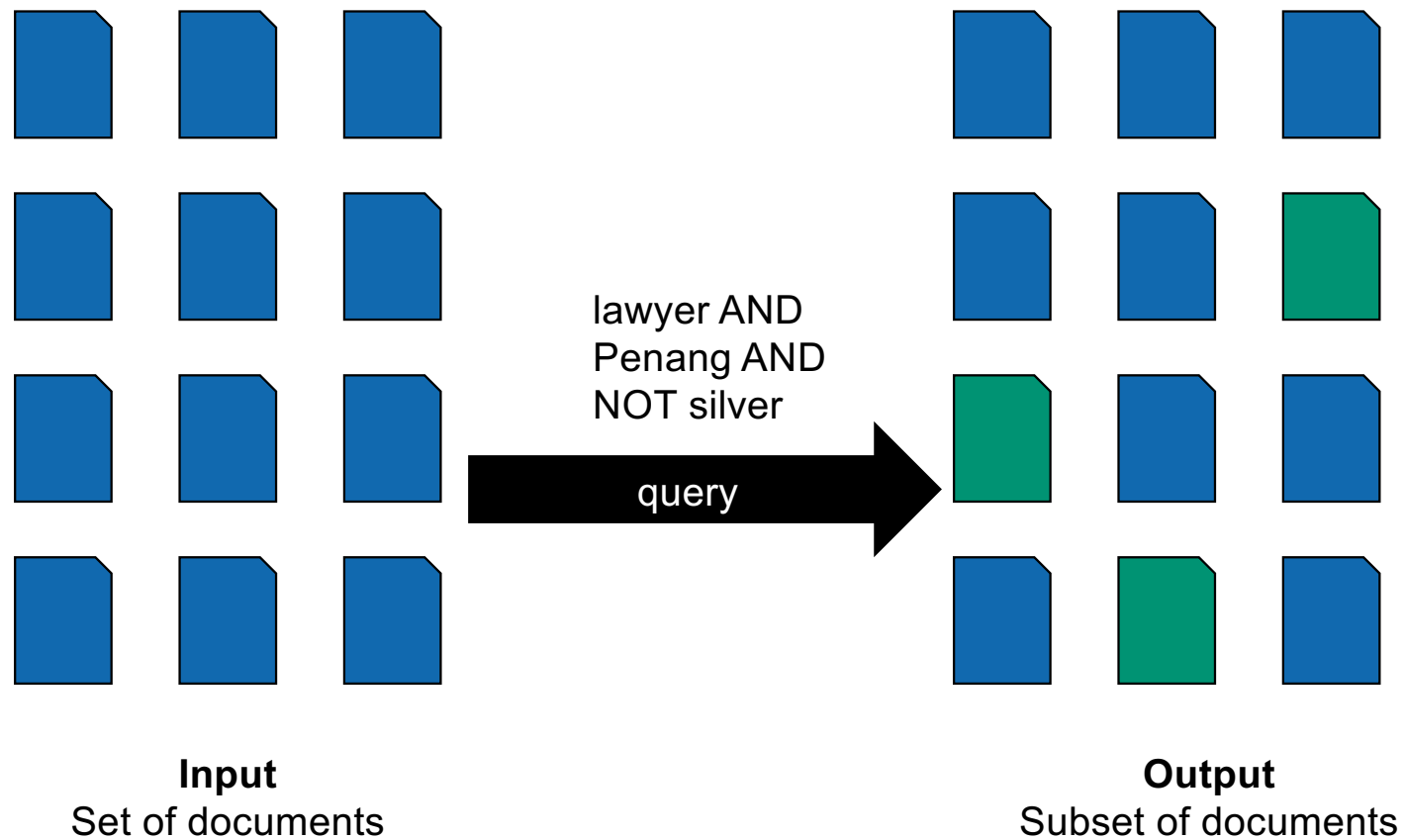


What we have seen so far

Warm up

a	→	6	1	2	3	5	6	8
b	→	5	3	4	7	8	9	
c	→	5	1	2	4	5	7	
d	→	5	1	3	5	8	9	
e	→	4	2	3	4	7		
f	→	6	1	2	4	5	8	9
g	→	4	3	5	7	8		

Boolean retrieval



Simple boolean language (EBNF)

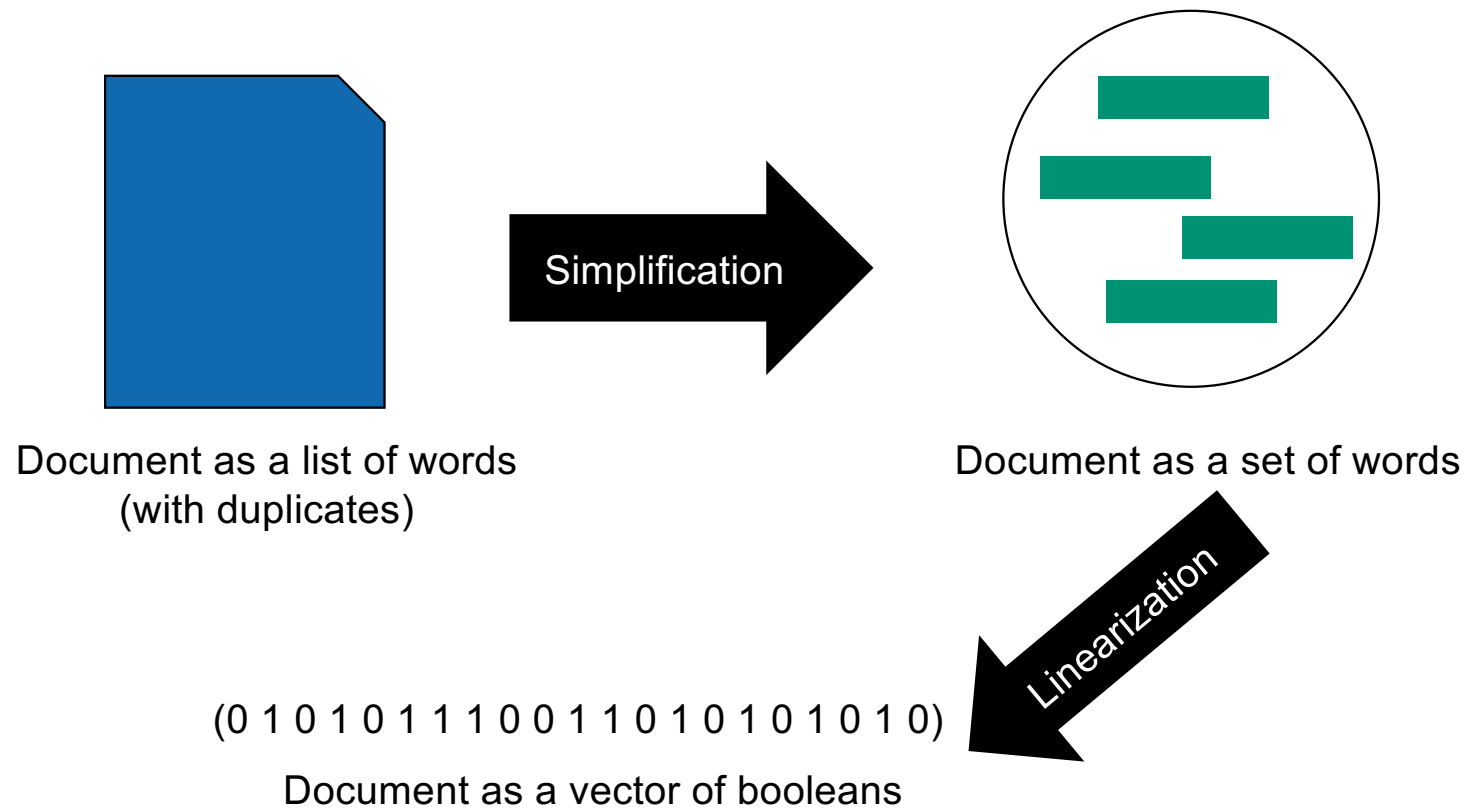
`PrimaryExpr ::= Term | "(" Expr ")"`

`NotExpr ::= "NOT"? PrimaryExpr`

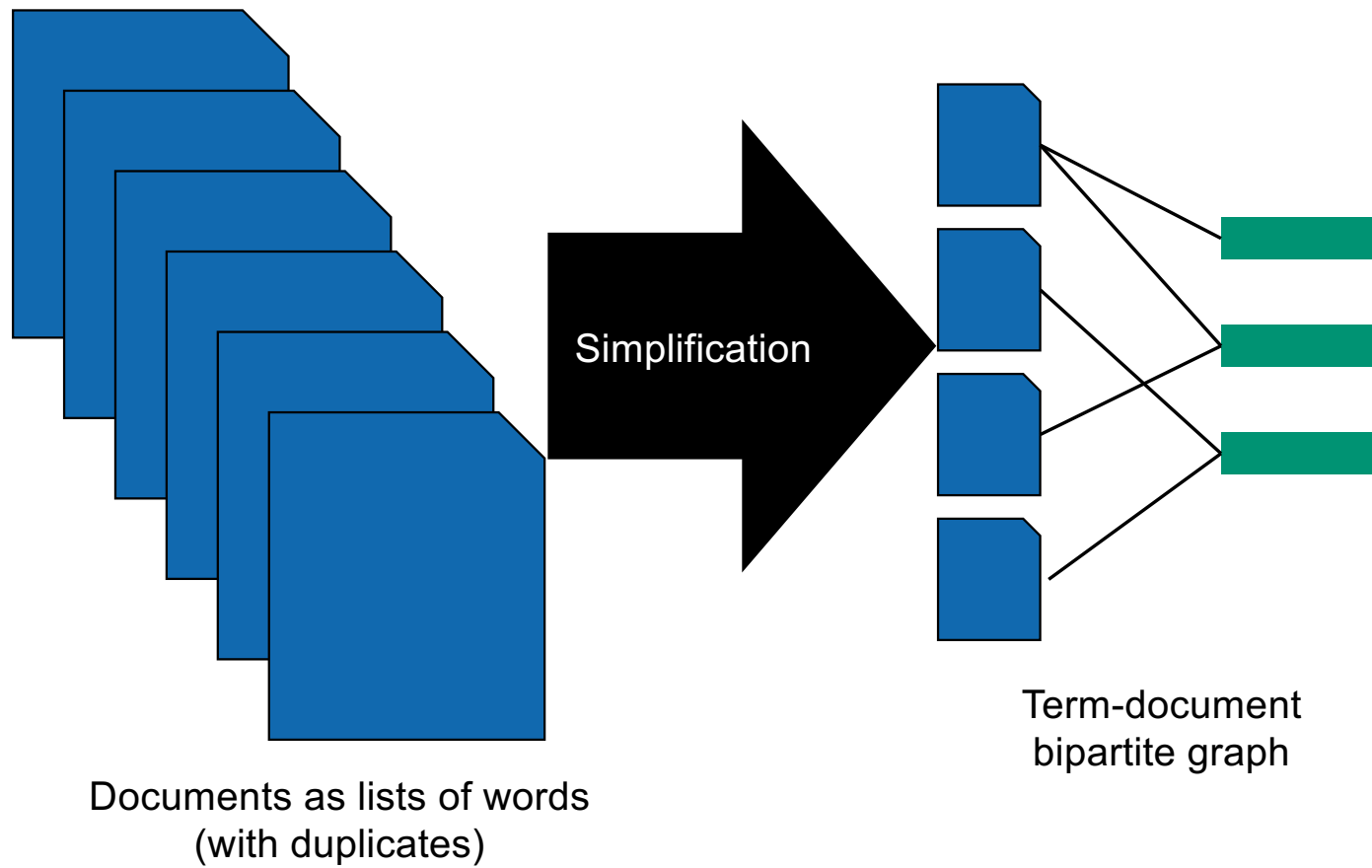
`AndExpr ::= NotExpr ("AND" NotExpr)*`

`Expr ::= NotExpr ("OR" AndExpr)*`

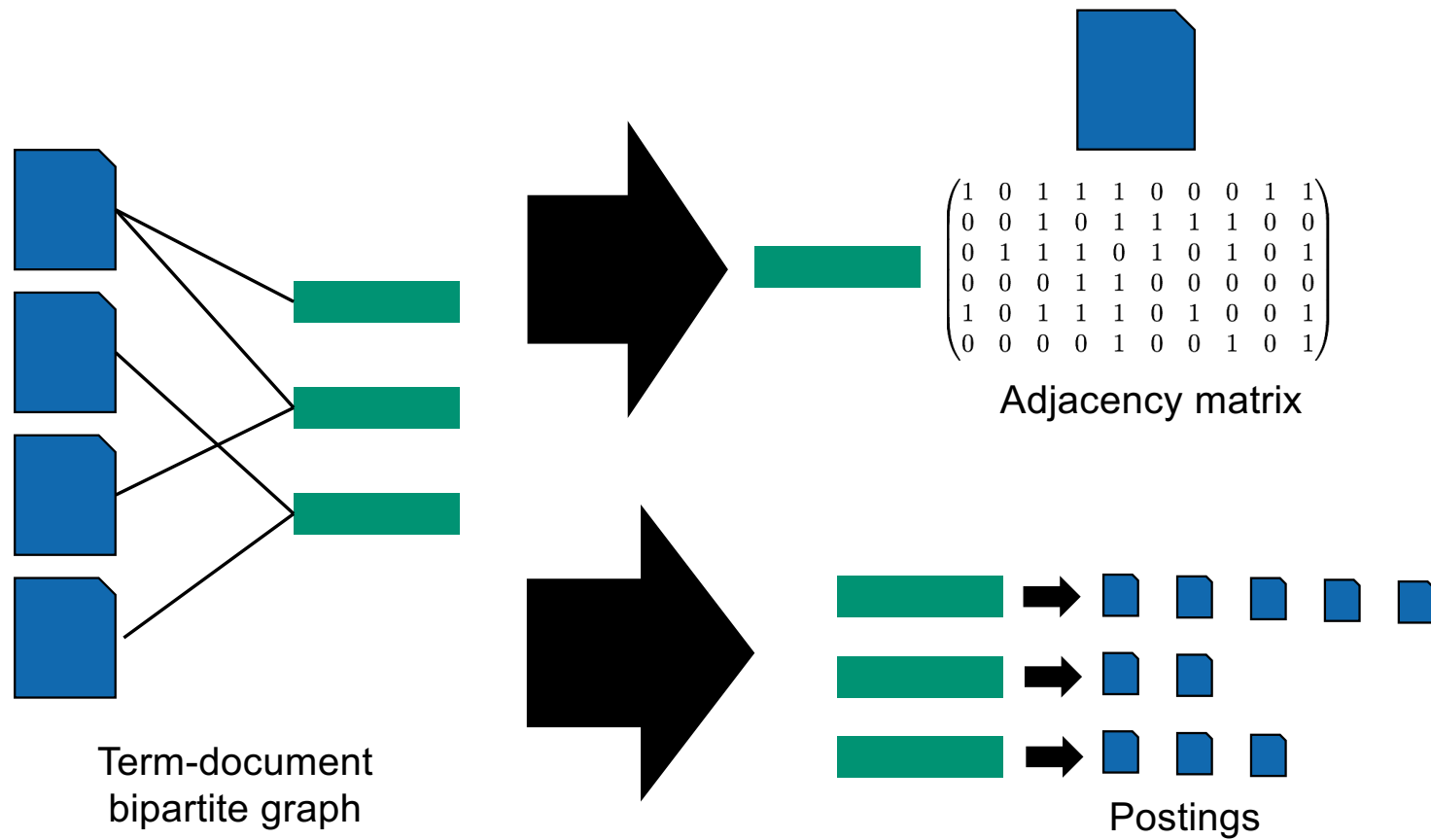
Model and abstraction



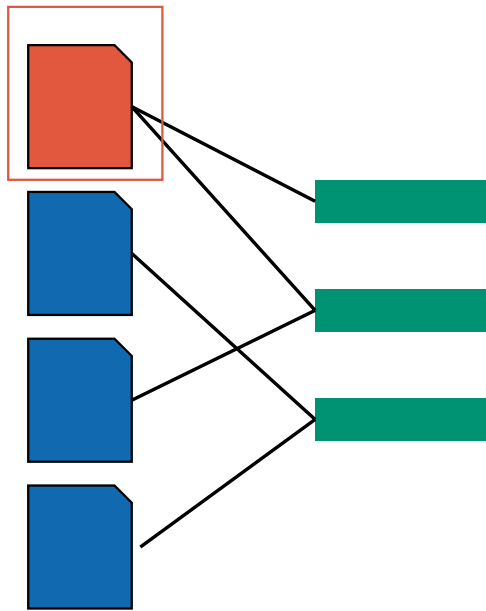
Term-document model



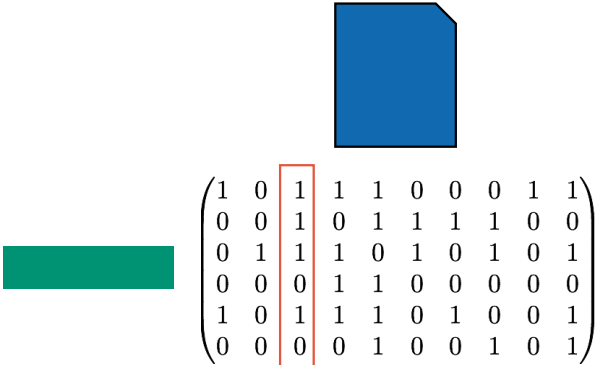
Term-document model



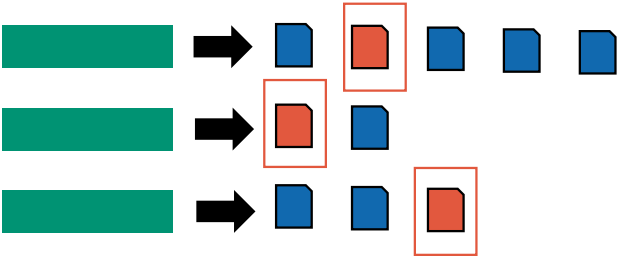
Document



Term-document bipartite graph

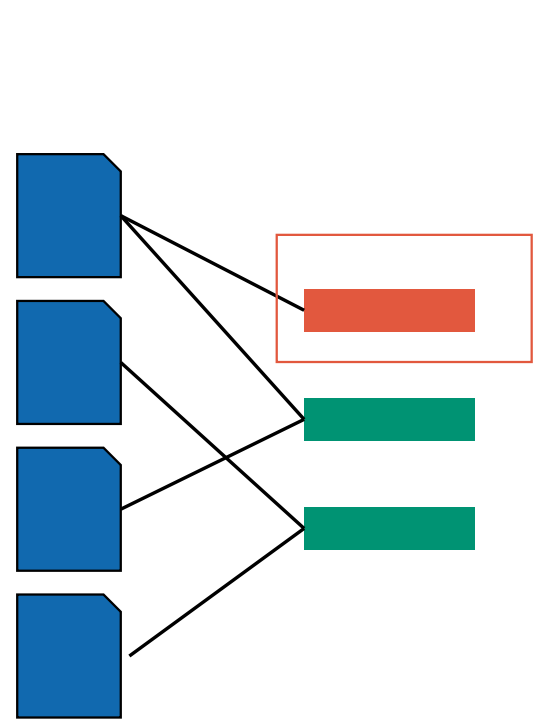


Adjacency matrix

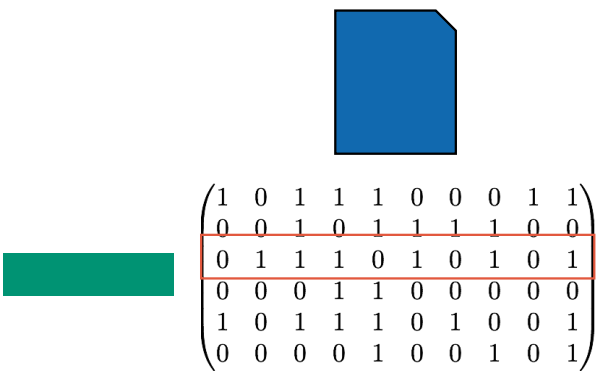


Postings

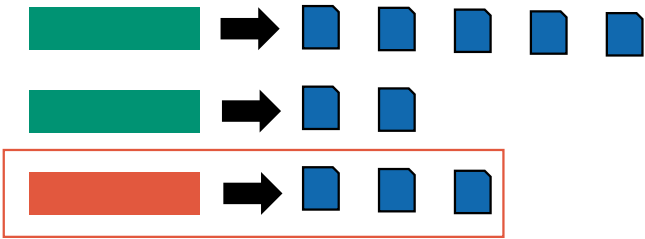
Term



Term-document bipartite graph

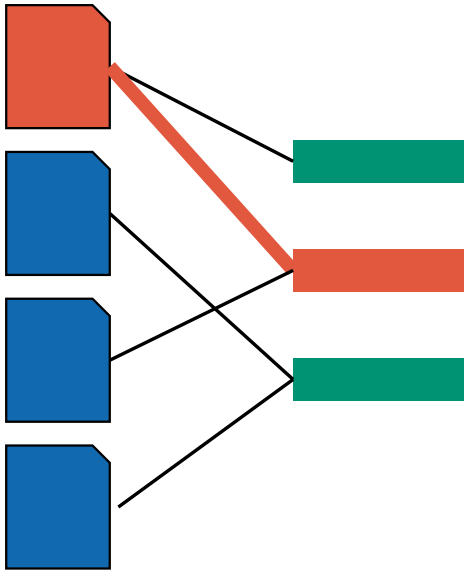


Adjacency matrix

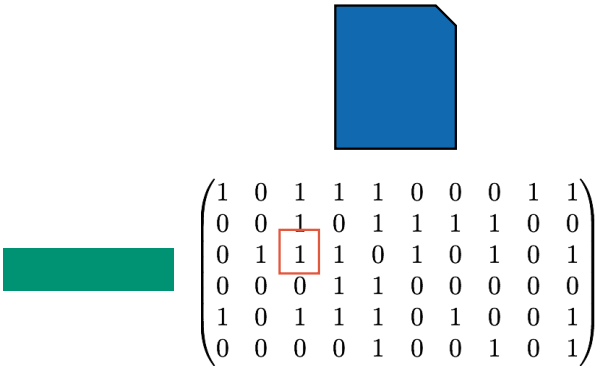


Postings

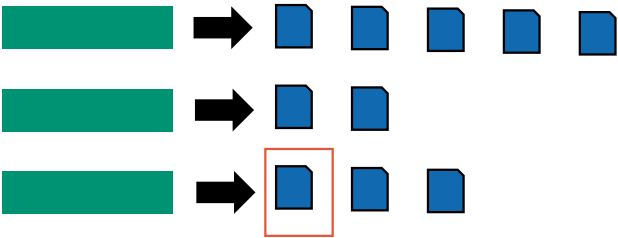
Posting



Term-document bipartite graph



Adjacency matrix



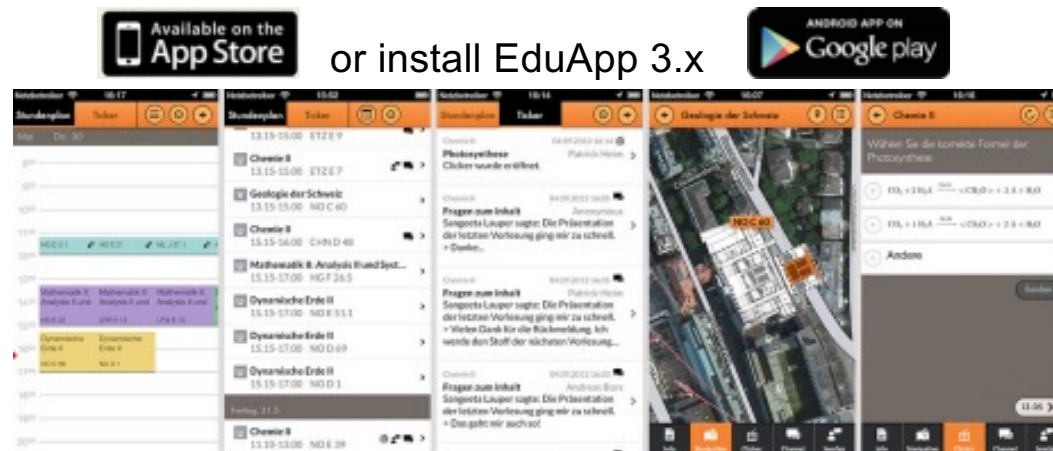
Postings

Index construction

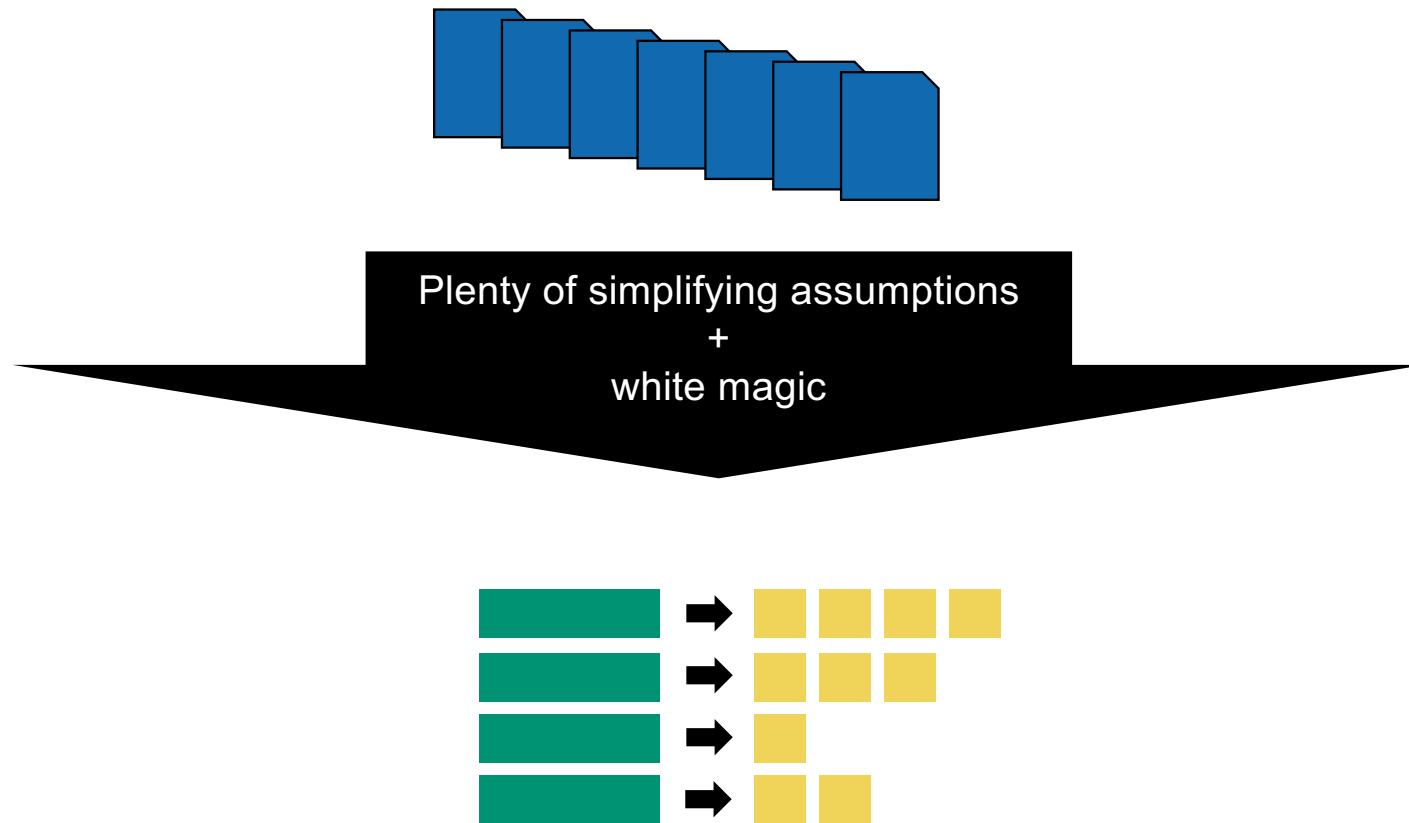
Poll

Go *now* to:

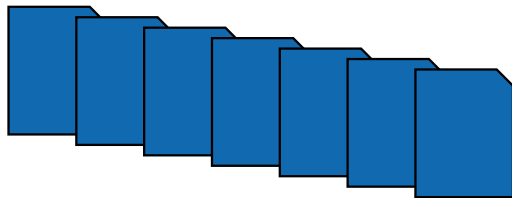
<https://eduapp-app1.ethz.ch/>



Last time...

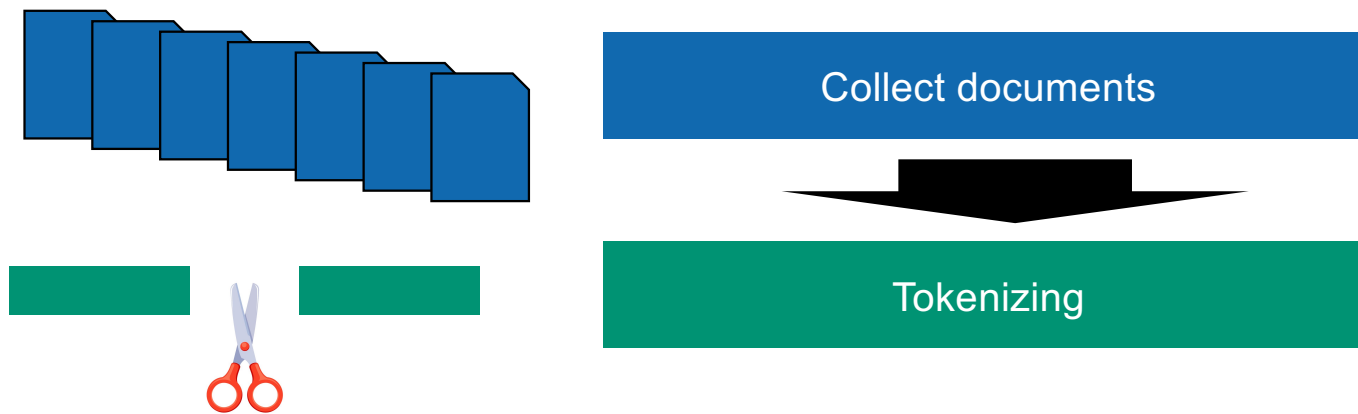


Index construction in reality...

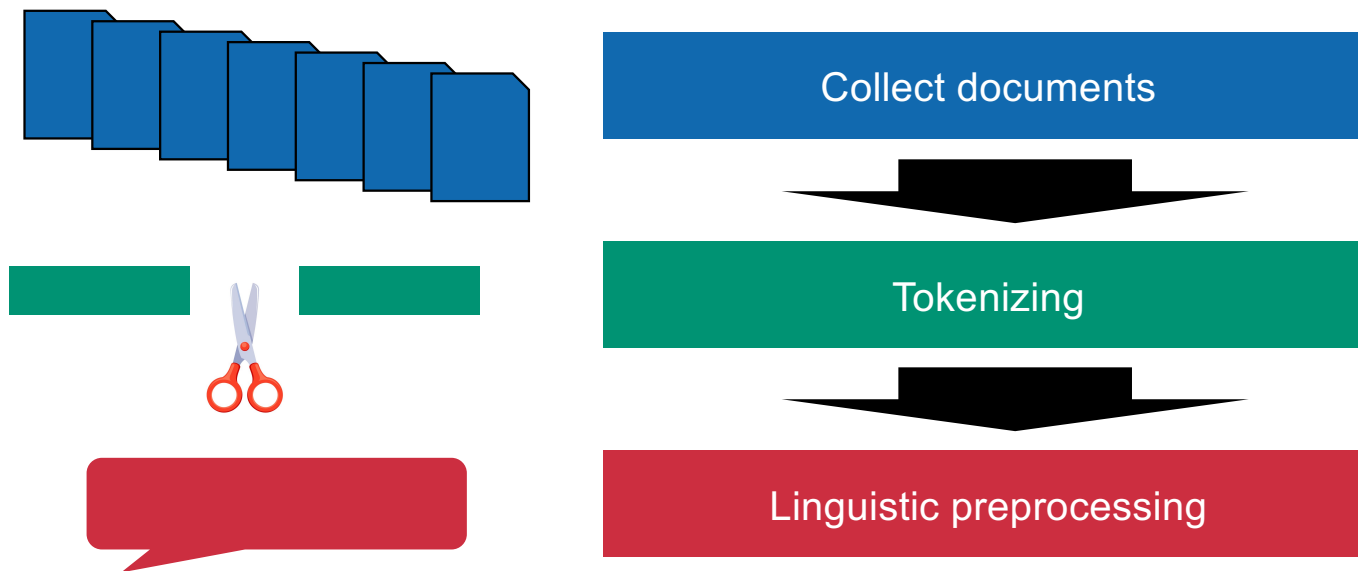


Collect documents

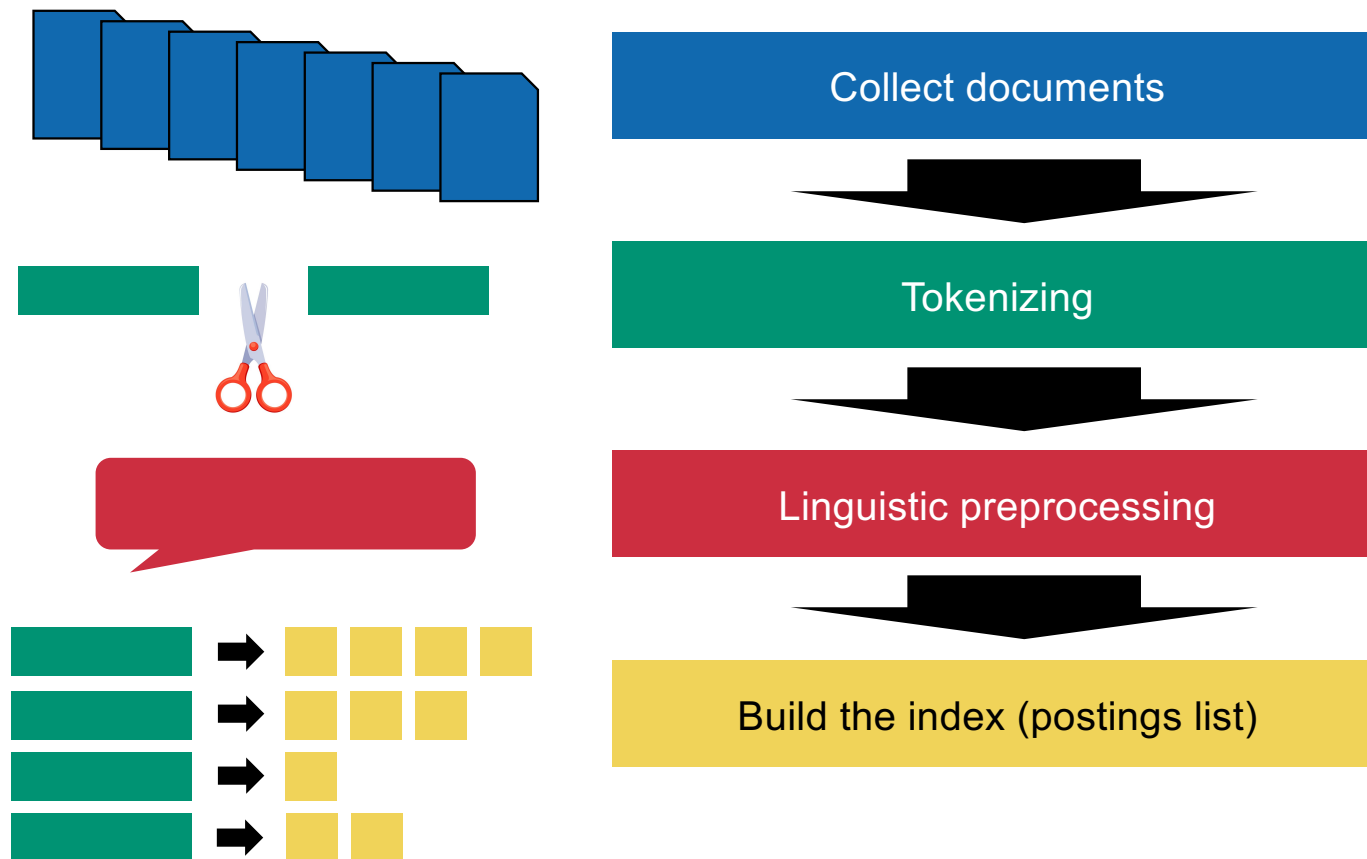
Index construction in reality...



Index construction in reality...



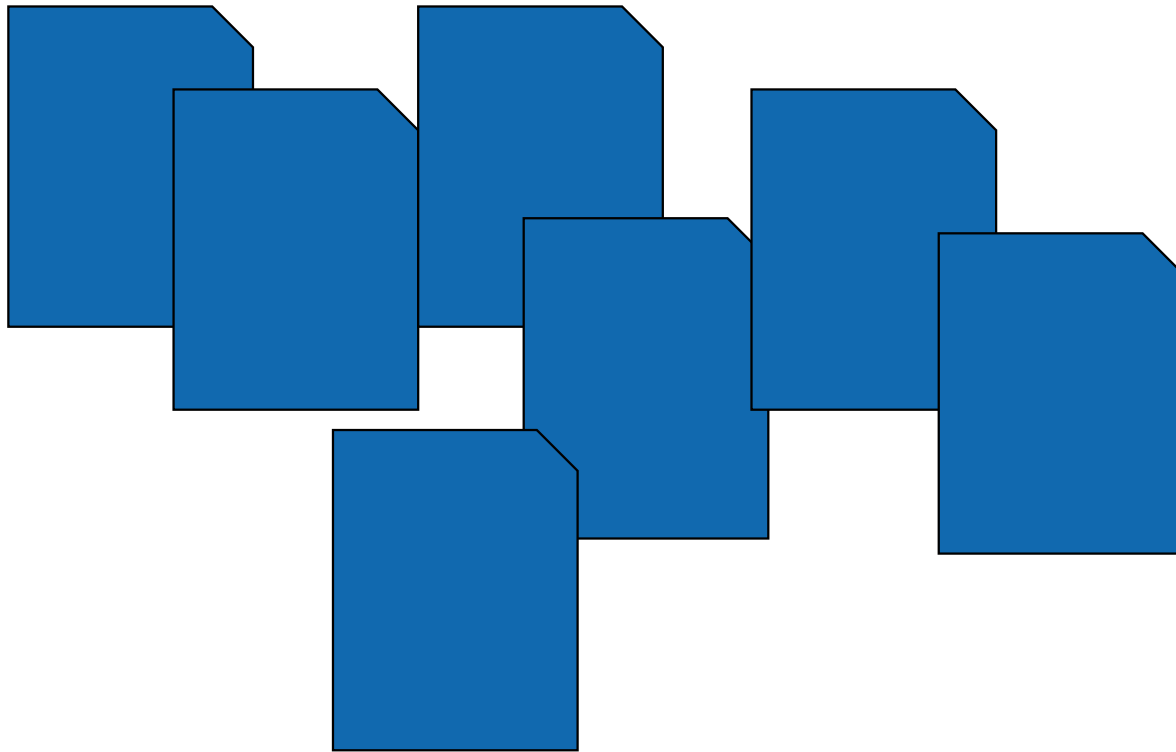
Index construction in reality...





Documents

Collecting documents



Collecting documents: first challenge



Collecting documents: first challenge

Possess it merely. That it should come to this!

Collecting documents: sequence of characters

Possess it merely. That it should come to this!



P	o	s	s	e	s	s		i	t		m	e	r	e	l	y	.		T	h	a	t		i	t		s	h	o	u
---	---	---	---	---	---	---	--	---	---	--	---	---	---	---	---	---	---	--	---	---	---	---	--	---	---	--	---	---	---	---

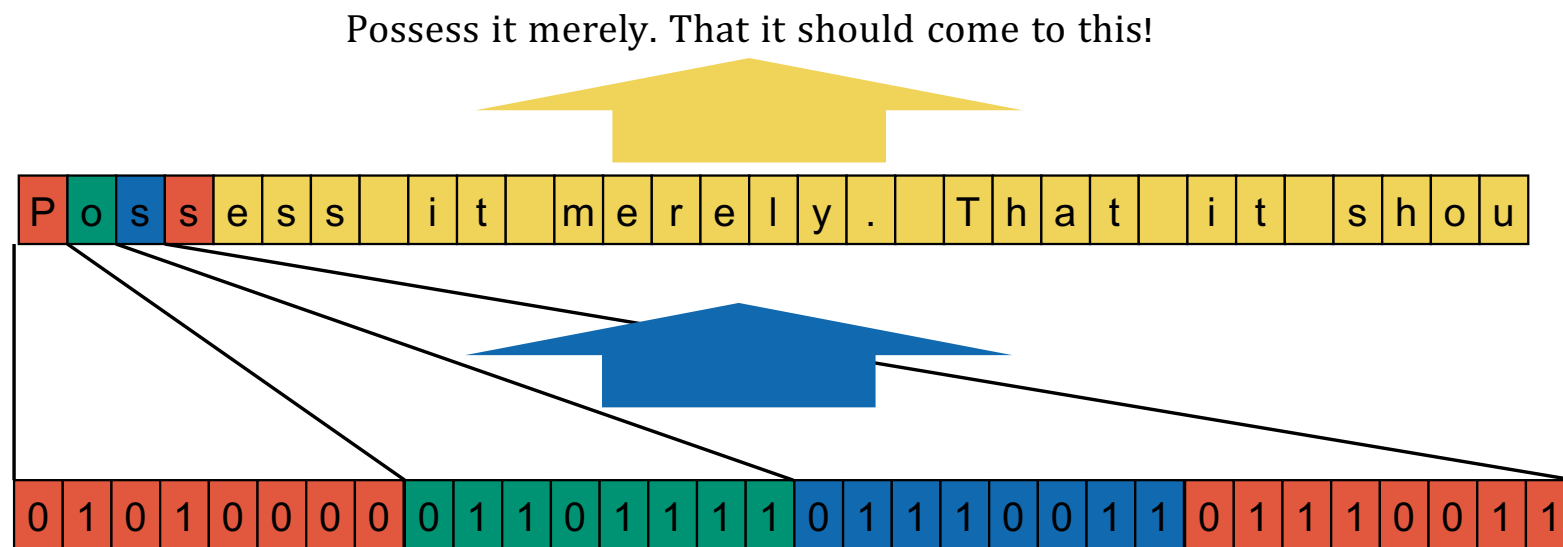
d		c	o	m	e		t	o		t	h	i	s	!
---	--	---	---	---	---	--	---	---	--	---	---	---	---	---

Character set

P o s s e s s i t m e r e l y . T h a t i t s h o u

d c o m e t o t h i s !

Collecting documents: encoding

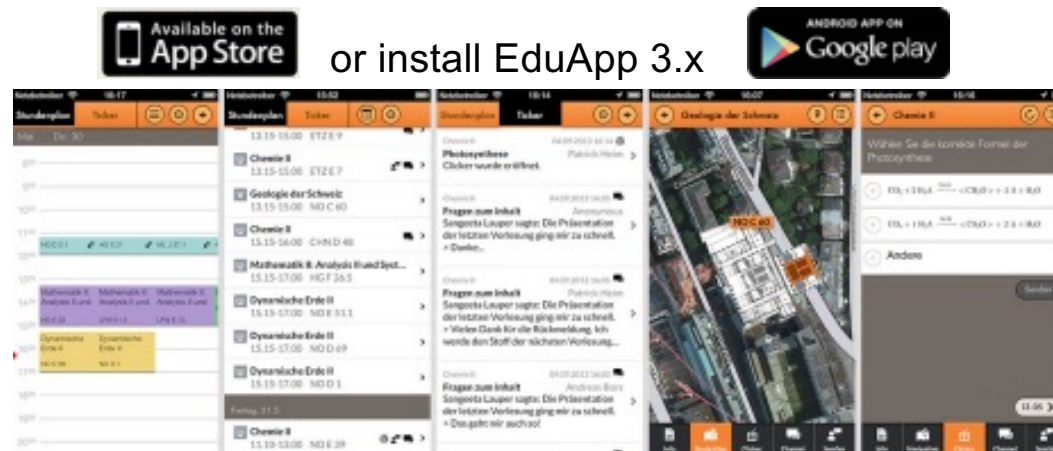


Here: ASCII

Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>



ASCII Table

	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9	.A	.B	.C	.D	.E	.F
0.	Control characters															
1.	Control characters															
2.	SP	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3.	0	1	2	3	4	5	6	7	8	9	:	:	<	=	>	?
4.	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5.	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6.	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7.	p	q	r	s	t	u	v	w	x	y	z	{		}	~	DEL

ASCII Table

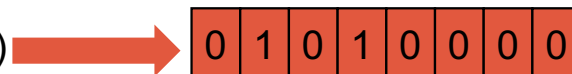
	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9	.A	.B	.C	.D	.E	.F
0.	Control characters															
1.	Control characters															
2.	SP	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3.	0	1	2	3	4	5	6	7	8	9	:	:	<	=	>	?
4.	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5.	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6.	↓	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7.	p	q	r	s	t	u	v	w	x	y	z	{		}	~	DEL

50 (hex)

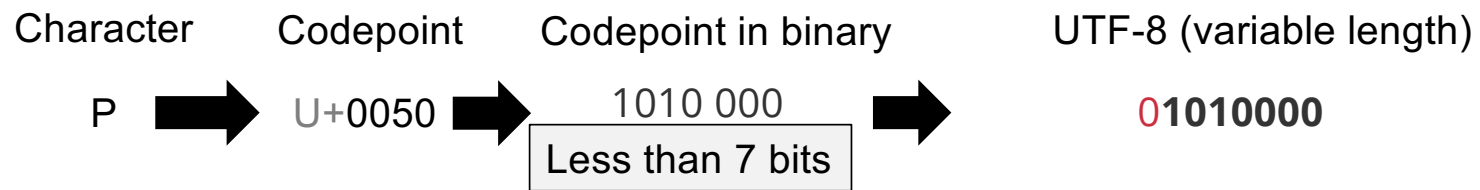
ASCII Table

	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9	.A	.B	.C	.D	.E	.F
0.	Control characters															
1.	Control characters															
2.	SP	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3.	0	1	2	3	4	5	6	7	8	9	:	:	<	=	>	?
4.	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5.	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6.	↓	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7.	p	q	r	s	t	u	v	w	x	y	z	{		}	~	DEL

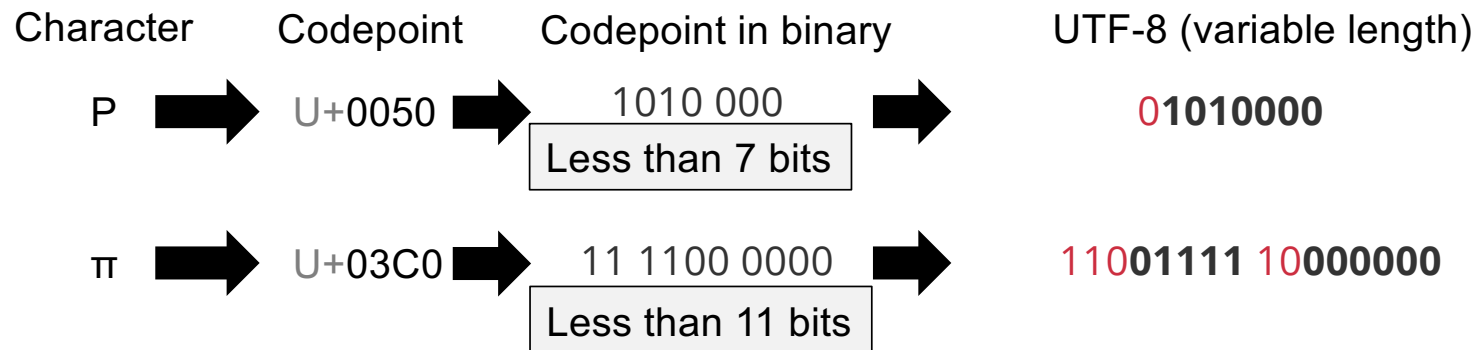
50 (hex)



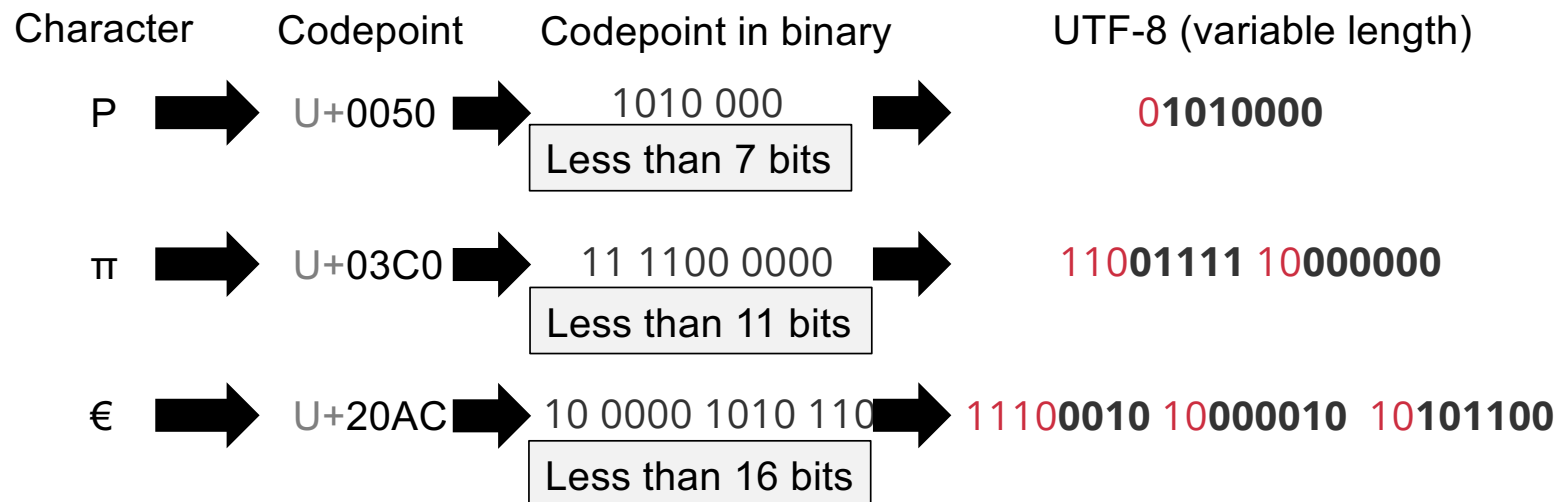
UTF-8



UTF-8



UTF-8



Collecting documents: first challenge



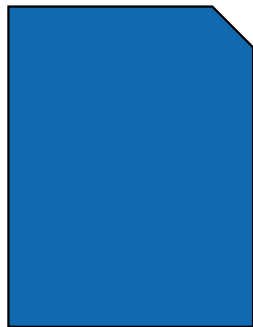
ASCII?

UTF-8?

ISO-Latin-1?

...?

Collecting documents: first challenge



User-defined

Annotated in document

Machine learning

...

Collecting documents: first challenge



Software-specific encoding

Collecting documents: first challenge



Data-specific encoding

&

<

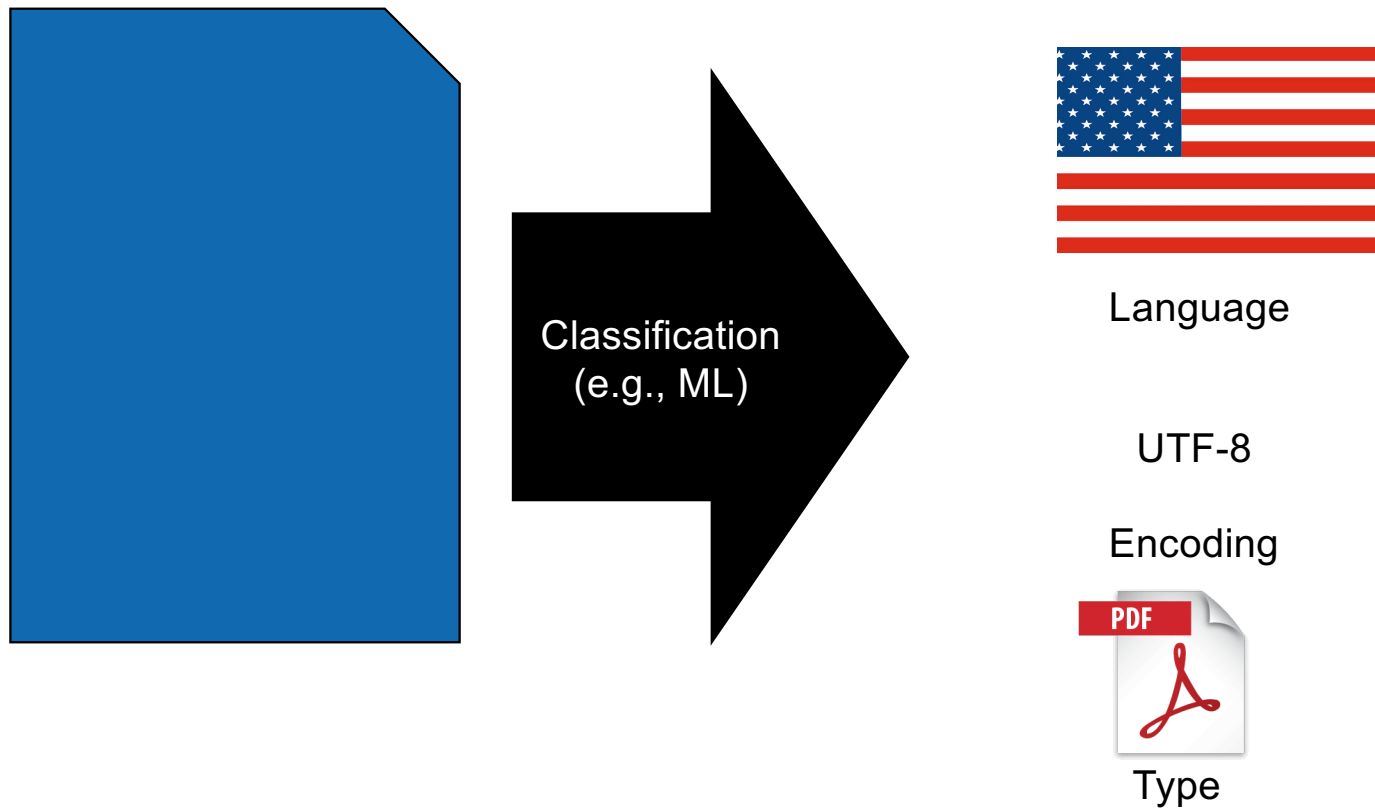
...

Collecting documents: first challenge

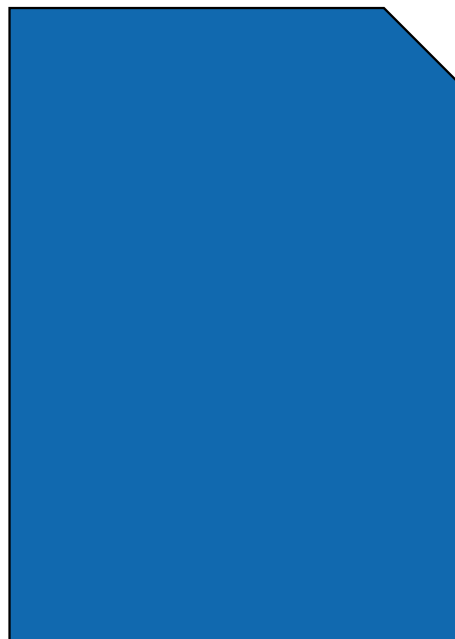


Binary encoding

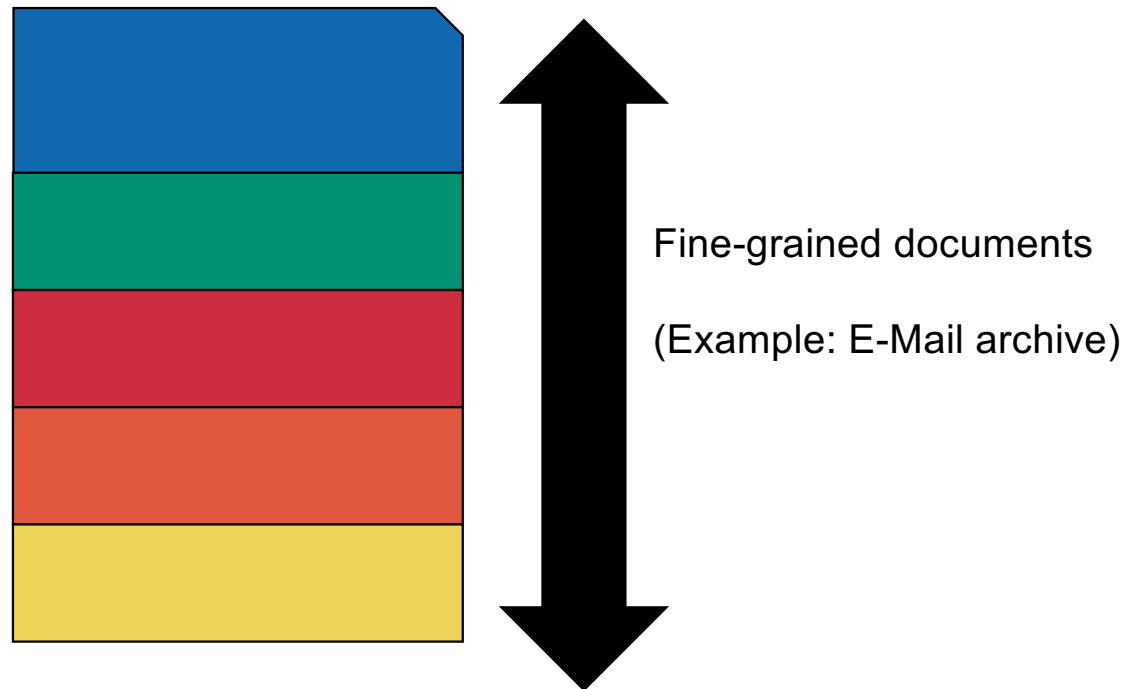
Collecting documents: first challenge



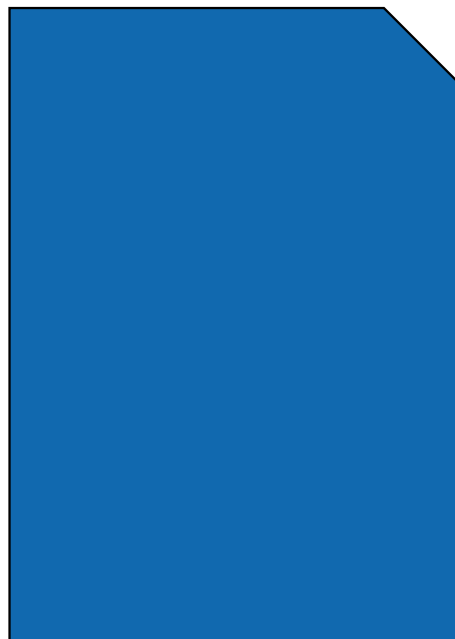
Collecting documents: second challenge



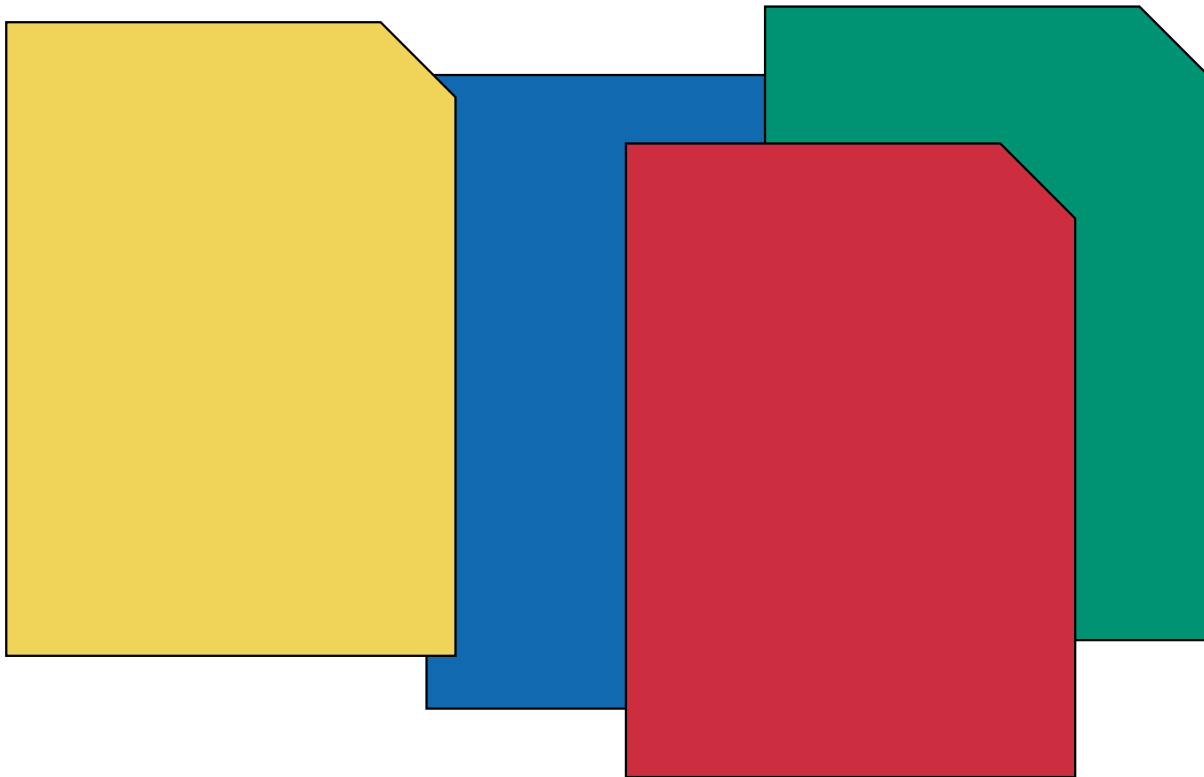
Collecting documents: second challenge



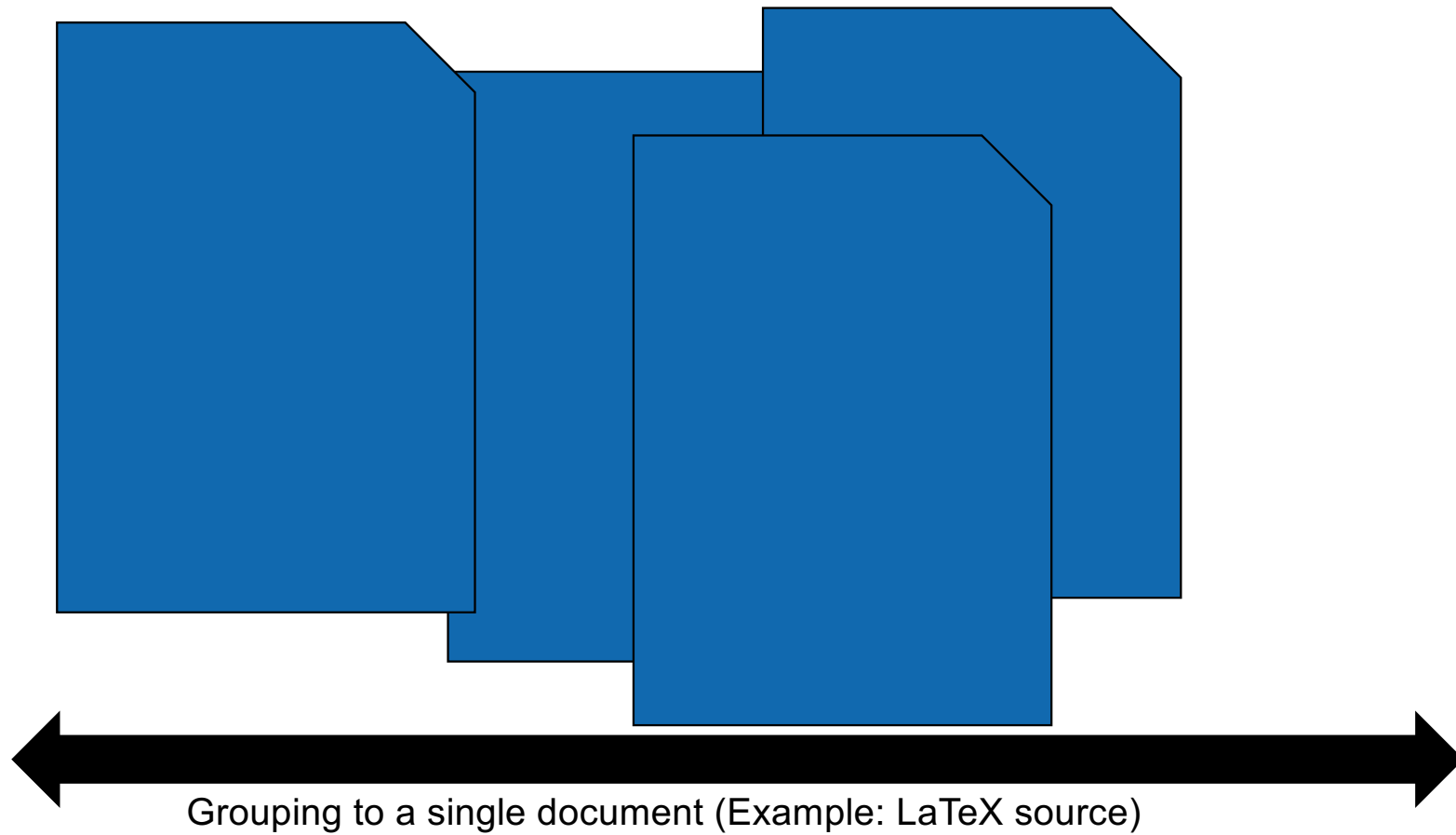
Collecting documents: second challenge



Collecting documents: second challenge



Collecting documents: second challenge





Term Vocabulary

Building an inverted index

Document 1

You come most carefully upon your hour.

Document 2

Take thy fair hour, Laertes; time be thine,

Document 3

My hour is almost come,

Document 4

Possess it merely. That it should come to this!

Building an inverted index

Document 1

You come most carefully upon your hour

Document 2

Take thy fair hour Laertes time be thine

Document 3

My hour is almost come

Document 4

Possess it merely That it should come to this

Throw away
punctuation

Building an inverted index

This is not trivial!

name@example.com

isn't

Jake O'Neill

the learn-it-all-by-heart methodology

website vs. web site

Kanton Basel Stadt

Throw away
punctuation

Corner cases

Hewlett-Packard

State-of-the-art

co-education

the hold-him-back-and-drag-him-away

maneuver data base

San Francisco

Los Angeles-based company

cheap San Francisco-Los Angeles fares

York University vs. New York University

Credits: H. Schütze, LMU

Corner cases: numbers

3/20/91

20/3/91

Mar 20, 1991

B-52

100.2.86.144

(800) 234-2333

800.234.2333

Credits: H. Schütze, LMU

English

I **would** like a coffee
You **would** like a coffee
He **would** like a coffee
We **would** like a coffee
You **would** like a coffee
They **would** like a coffee

English

I **want** a coffee
You **want** a coffee
He **wants** a coffee
We **want** a coffee
You **want** a coffee
They **want** a coffee

Swedish

Jag **skulle vilja ha** en kaffe
Du **skulle vilja ha** en kaffe
Han **skulle vilja ha** en kaffe
Vi **skulle vilja ha** en kaffe
Ni **skulle vilja ha** en kaffe
De **skulle vilja ha** en kaffe

German

Ich **möchte** einen Kaffee
Du **möchtest** einen Kaffee
Er **möchte** einen Kaffee
Wir **möchten** einen Kaffee
Ihr **möchtet** einen Kaffee
Sie **möchten** einen Kaffee

German

Donaudampfschiffahrtselektrizitätenhauptbetriebswerkbauunterbeamtengesellschaft

Swiss German

I ha tänkt, du heggish en kaffee **welle trinke**



Chinese

中华人民共和国，简称中国，是位於东亚的社会主义国家，首都位于北京[12]。中国領土面積約960萬平方公里，是世界上純陸地[註 9]面積第二大、陸地[註 10]面積第三大、總面積第三大或第四大的國家[註 11][13]，當中劃分為23個省份[註 12]、5個自治區、4個直轄市和2個特別行政區。

Hindi

मैं इस टर्म अच्छा पढ़ रहा हूँ।

मैं इस टर्म अच्छा पढ़ रही हूँ।

मुझे इंफ़ॉर्मेशन रिट्रीवल बहुत पसंद है।

Hindi

Male speaker 
मैं इस टर्म अच्छा पढ़ रहा हूँ।
| raha 1st person
Oblique case मैं इस टर्म अच्छा पढ़ रही हूँ।
rahee Female speaker 

मुझे इंफ़र्मेशन रिट्रीवल बहुत पसंद है।
To me 3rd person

Hindi

मैं इस टर्म अच्छा पढ़ूँगा।

Male speaker



Future
1st person

मैं इस टर्म अच्छा पढ़ूँगी

Female speaker



Polysynthetic Languages
Siberian Yupik

neghyaghtughyugumayaghpetallu

**Source: Polysynthetic Language: Central Siberian
Yupik**
W J De Reuse

Polysynthetic Languages

Siberian Yupik

neghyaghtughyugumayaghpetallu
eat

Polysynthetic Languages

Siberian Yupik

neghyaghtughyugumayaghpetaaallu

eat go to

Polysynthetic Languages

Siberian Yupik

neghyaghtughyugumayaghpetaaallu

eat go to want

Polysynthetic Languages

Siberian Yupik

Past
neghyaghtughyugumayaghpetallu
eat go to want

Polysynthetic Languages

Siberian Yupik

<Frustration>

Past

neghyaghtughyugumayaghpetaaallu

eat go to want

Polysynthetic Languages

Siberian Yupik

<Frustration>

Past

neghyaghtughyugumayaghpetaaallu

eat go to want

inferential
evidential
(turns out)

Polysynthetic Languages

Siberian Yupik

<Frustration>

Past

3rd/3rd

neghyaghtughyugumayaghpetallu

eat go to want

inferential
evidential
(turns out)

Polysynthetic Languages

Siberian Yupik

<Frustration>

Past

3rd/3rd

neghyaghtughyugumayaghpetallu

eat

go to

want

also

inferential
evidential
(turns out)

Polysynthetic Languages

Siberian Yupik

neghyaghtughyugumayaghpetallu

Also, it turns out she/he wanted to go eat it, but...

Tokenize

You come most carefully upon your hour

Take fair thy hour Laertes time be thine

My hour is almost come

Possess it merely That it should come to this

Token

You come most carefully upon your hour

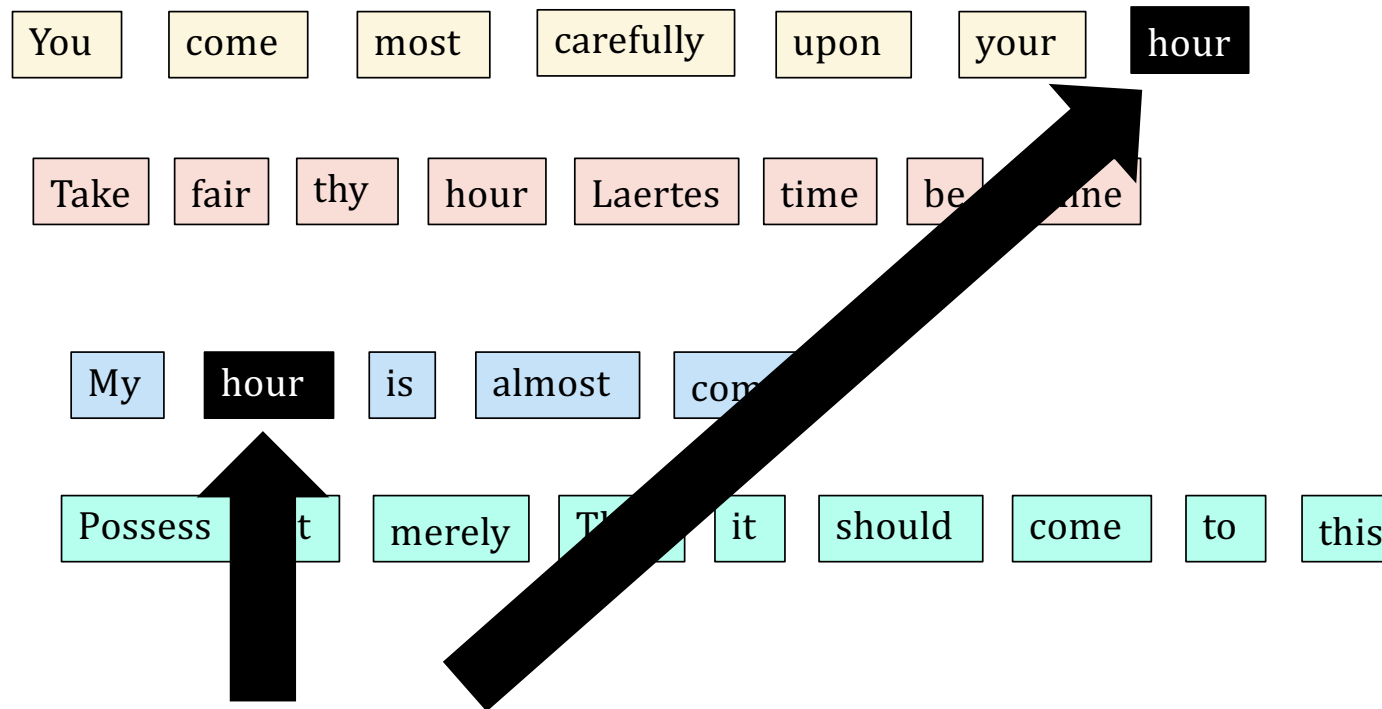
Take fair thy hour Laertes time be thine

My hour is almost come

Possess it merely That it should come to this

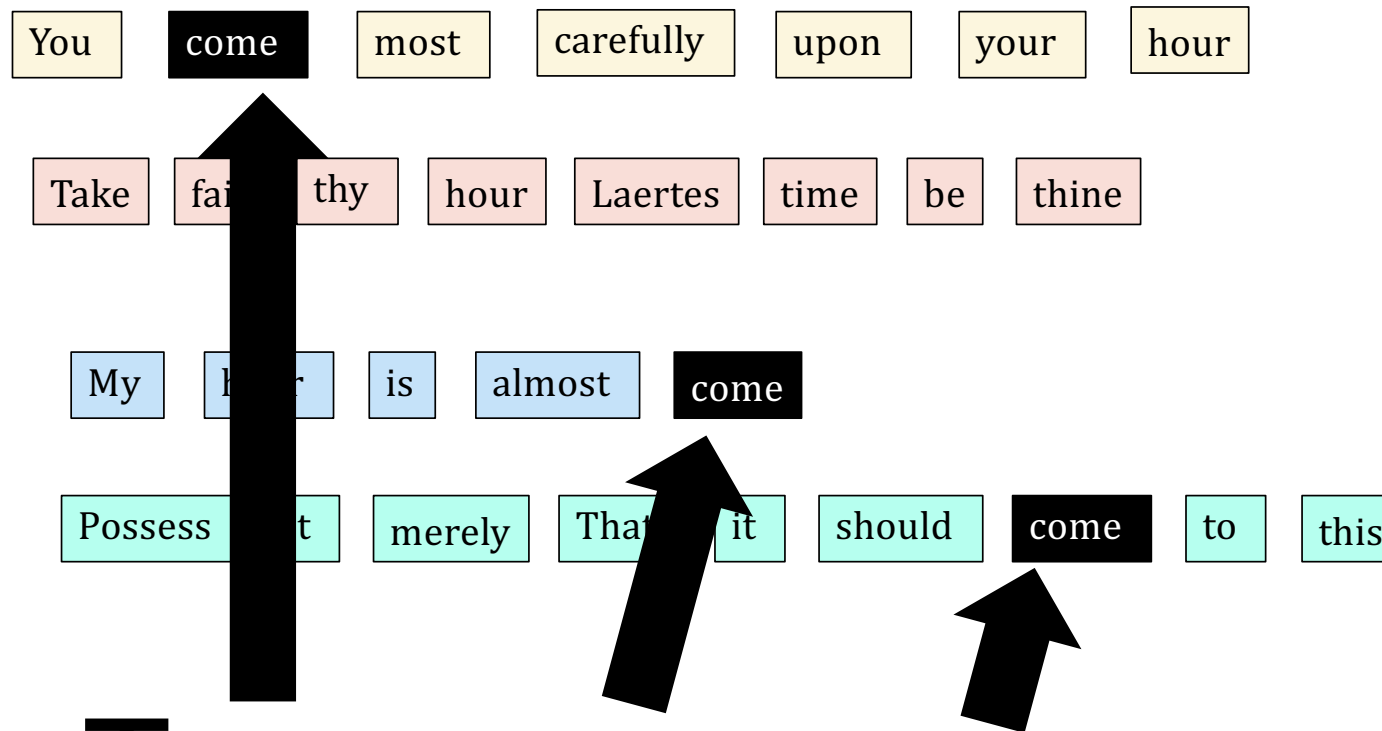
Token = grouped sequence of characters

Type



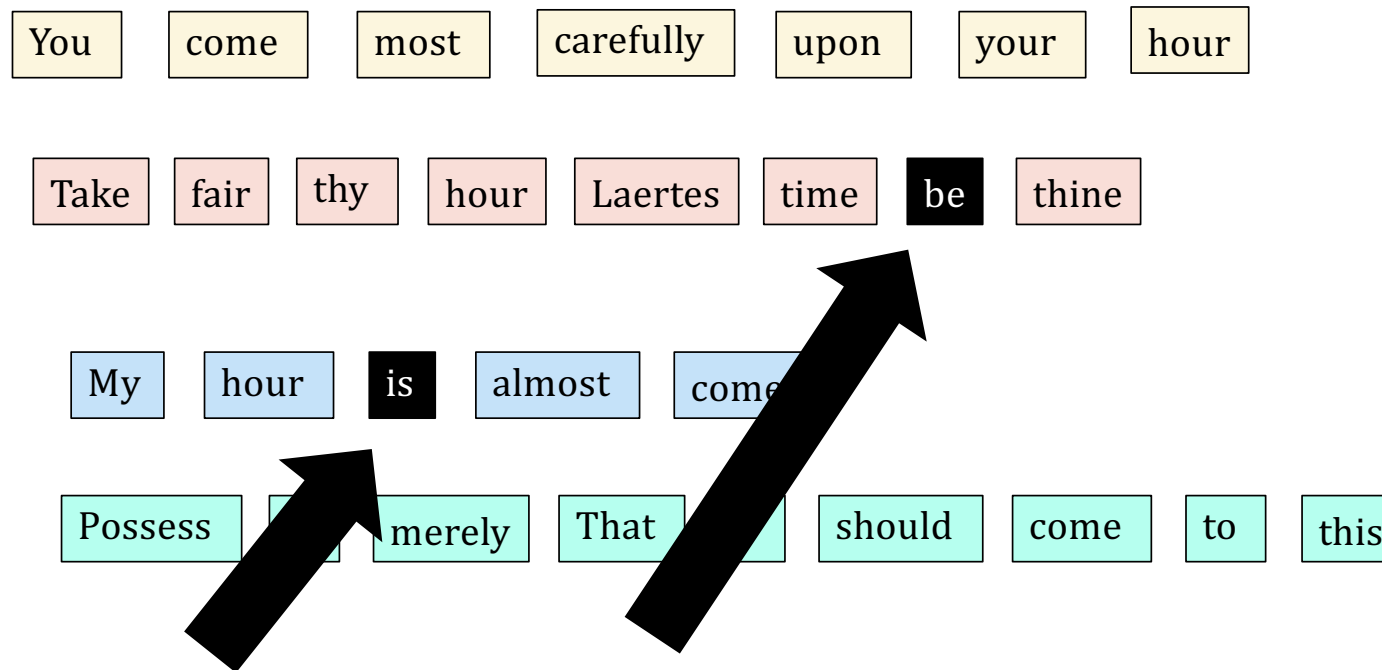
Type = equivalence class (same sequences)

Type



Type = equivalence class (same sequences)

Type



Type = equivalence class (same sequences)

Stop words

You come most carefully upon your hour

Take fair thy hour Laertes time be thine

My hour is almost come

Possess it merely That it should come to this

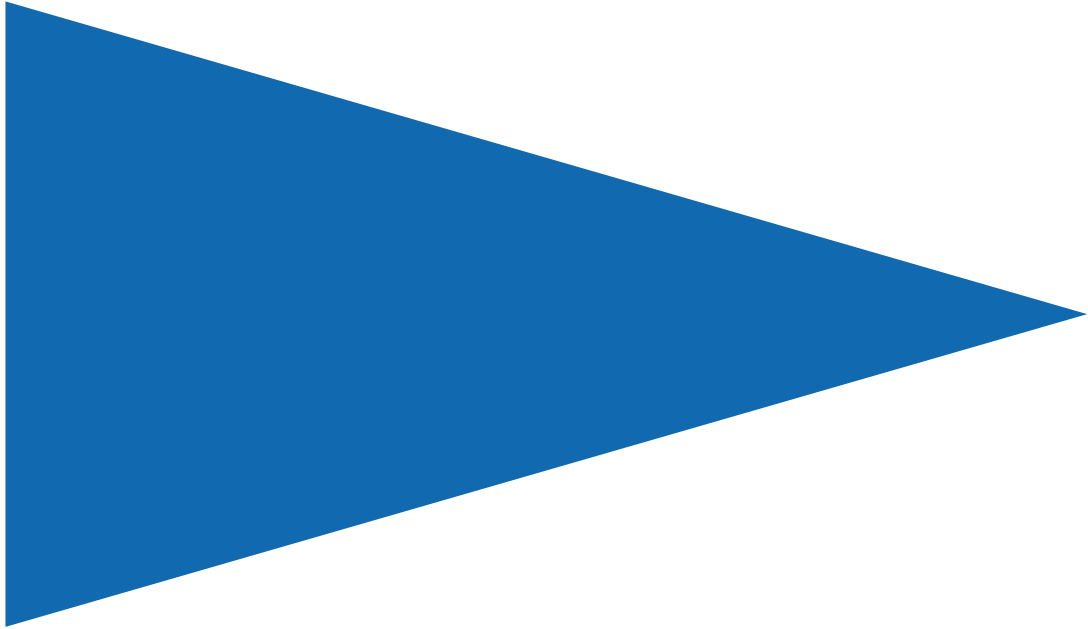
Reuters's list

a
an
and
are
as
at
be
by
for
from
has
he
in

is
it
its
of
on
that
the
to
was
were
will
with

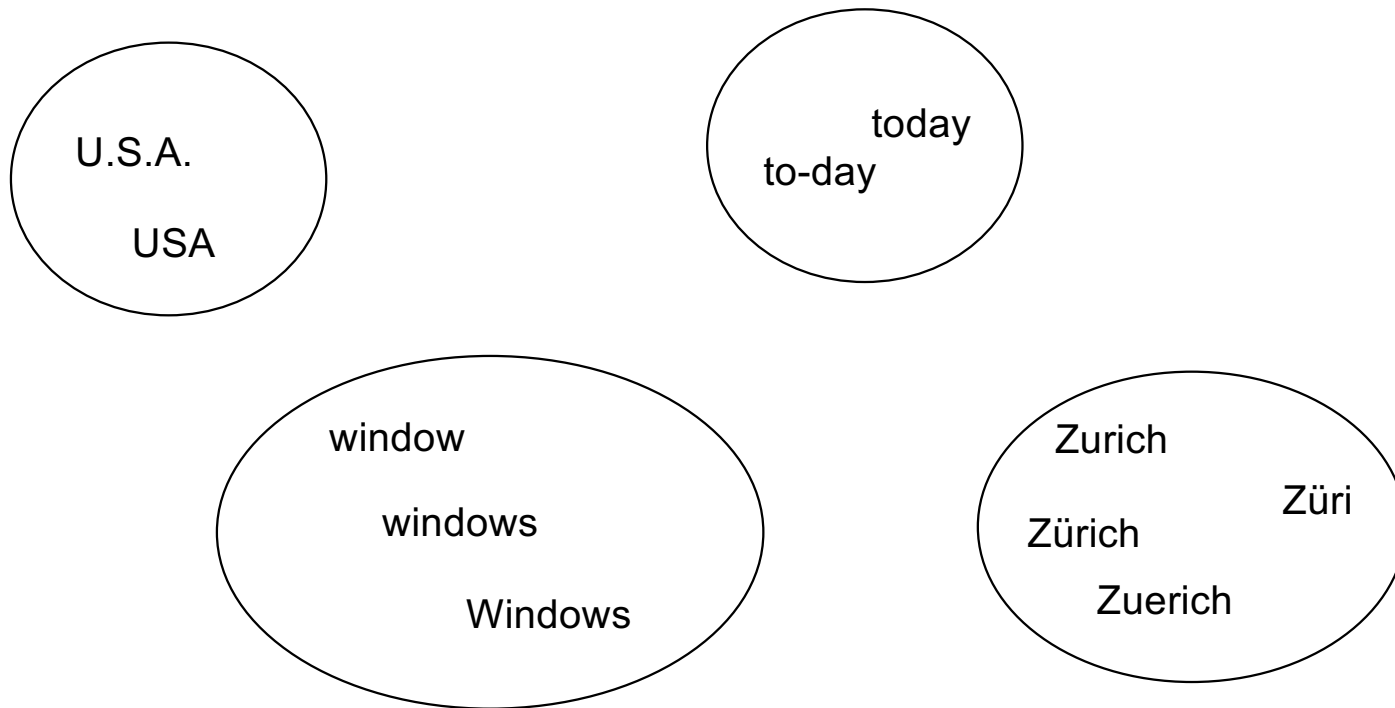
Trend

Large Isit (200-300)



No list at all

Equivalence classes of tokens (types)



Normalization

U.S.A.  USA
to-day  today

Rules that remove characters are the easy part.

Expansion (rather than equivalence classes)

Introduces asymmetry!

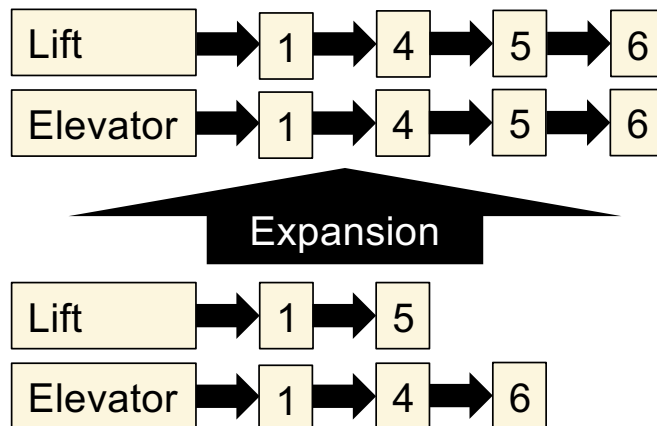
Windows → Windows

windows → windows, Windows, window

window → window, windows

When to expand?

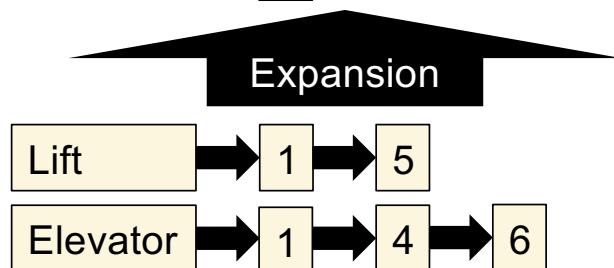
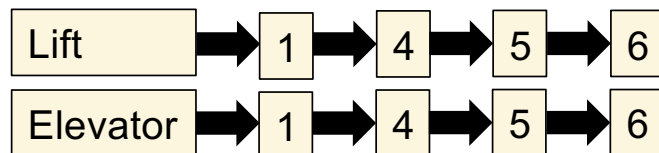
Upon indexing



When to expand?

Upon indexing

Lift |

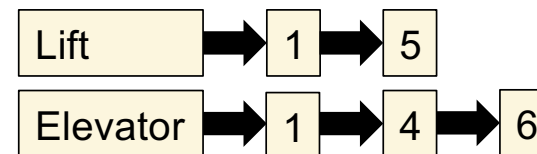


Upon querying

Lift |



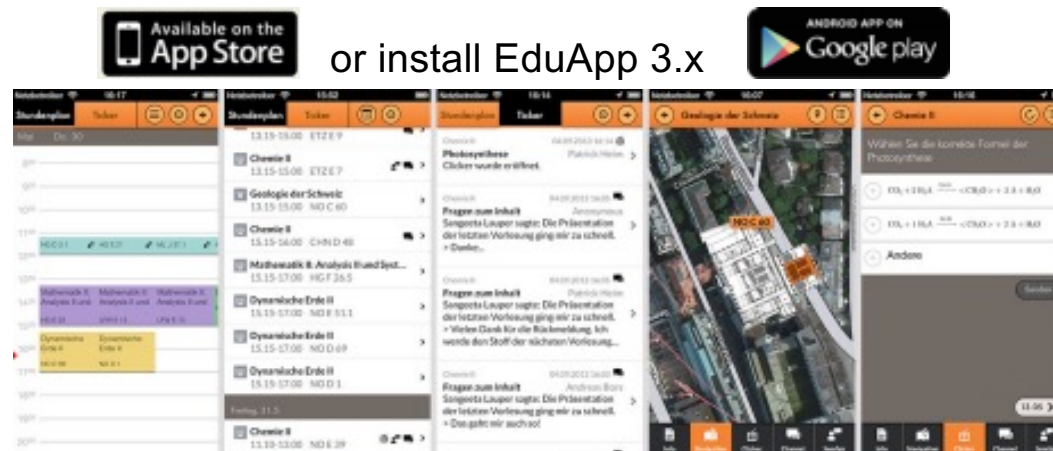
Lift OR Elevator



Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>



Normalization: accents and diacritics

cliché → cliche

Zürich → Zuerich

Normalization: lowercasing

C.A.T  cat

Schweiz  schweiz

Normalization: truecasing

The company Apple has launched a new iProduct.



Apple

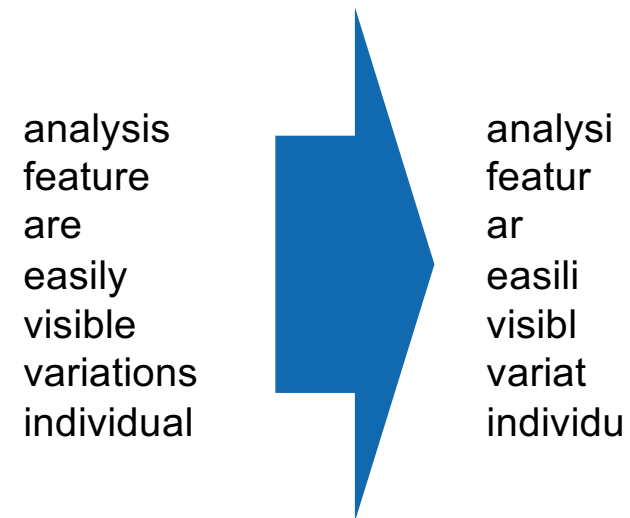
Machine learning inside

apples



Apples fall from trees.

Stemming



"Chop" letters of the word

Porter Stemmer

(m>0) ENCI -> ENCE	valenci -> valence
(m>0) ANCI -> ANCE	hesitanci -> hesitance
(m>0) IZER -> IZE	digitizer -> digitize
(m>0) ABLI -> ABLE	conformabli -> conformable
(m>0) ALLI -> AL	radicalli -> radical
(m>0) ENTLI -> ENT	differentli -> different
(m>0) ELI -> E	vileli -> vile
(m>0) OUSLI -> OUS	analogousli -> analogous
(m>0) IZATION -> IZE	vietnamization -> vietnamize
(m>0) ATION -> ATE	predication -> predicate
(m>0) ATOR -> ATE	operator -> operate
(m>0) ALISM -> AL	feudalism -> feudal
(m>0) IVENESS -> IVE	decisiveness -> decisive
(m>0) FULNESS -> FUL	hopefulness -> hopeful
(m>0) OUSNESS -> OUS	callousness -> callous

<https://tartarus.org/martin/PorterStemmer/>

Porter Stemmer

Such an analysis can reveal features that are not easily visible from the variations in the individual genes and can lead to a picture of expression that is more biologically transparent and accessible to interpretation

Porter

such an analysis can reveal features that are not easily visible from the variations in the individual genes and can lead to a picture of expression that is more biologically transparent and accessible to interpretation

Lovins

such an analysis can reveal features that are not easily visible from the variations in the individual genes and can lead to a picture of expression that is more biologically transparent and accessible to interpretation

Paice

such an analysis can reveal features that are not easily visible from the variations in the individual genes and can lead to a picture of expression that is more biologically transparent and accessible to interpretation

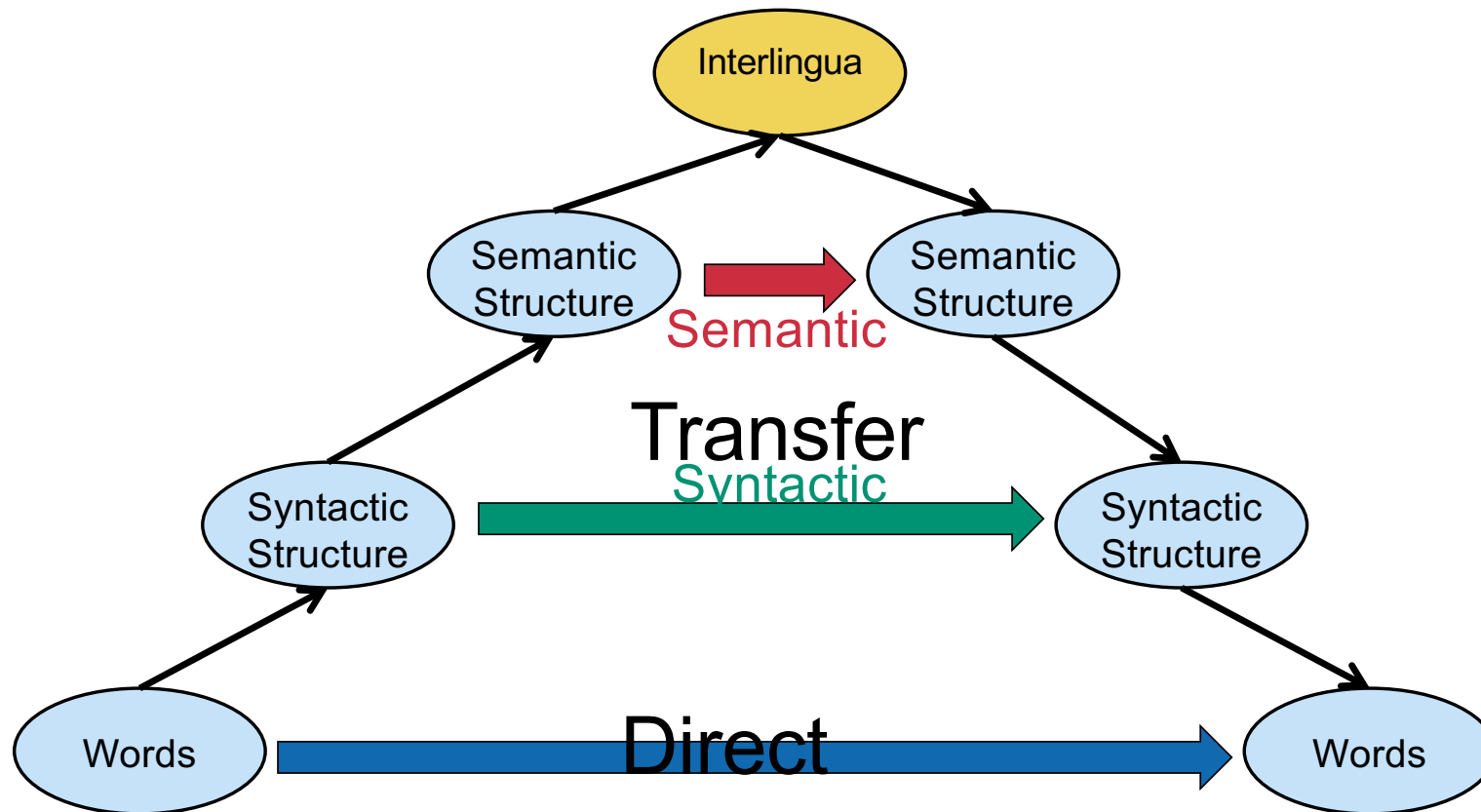
Source: the textbook (Introduction to Information Retrieval)

Lemmatization

Building equivalence classes or expanding with

Natural Language Processing

The Vauquois Triangle

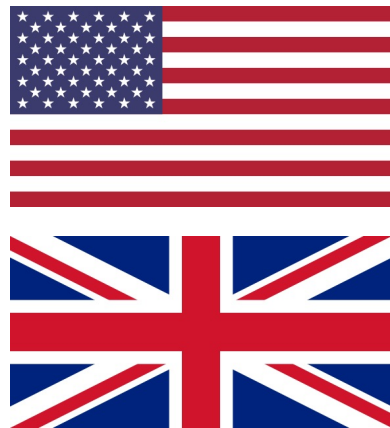


Lemmatization

Full morphological analysis

computer
compute
computes
computed
computation
computing
...

Lemmatization or Stemming?



Lemmatization does not help
and can even degrade performance
for English documents

Lemmatization or Stemming?



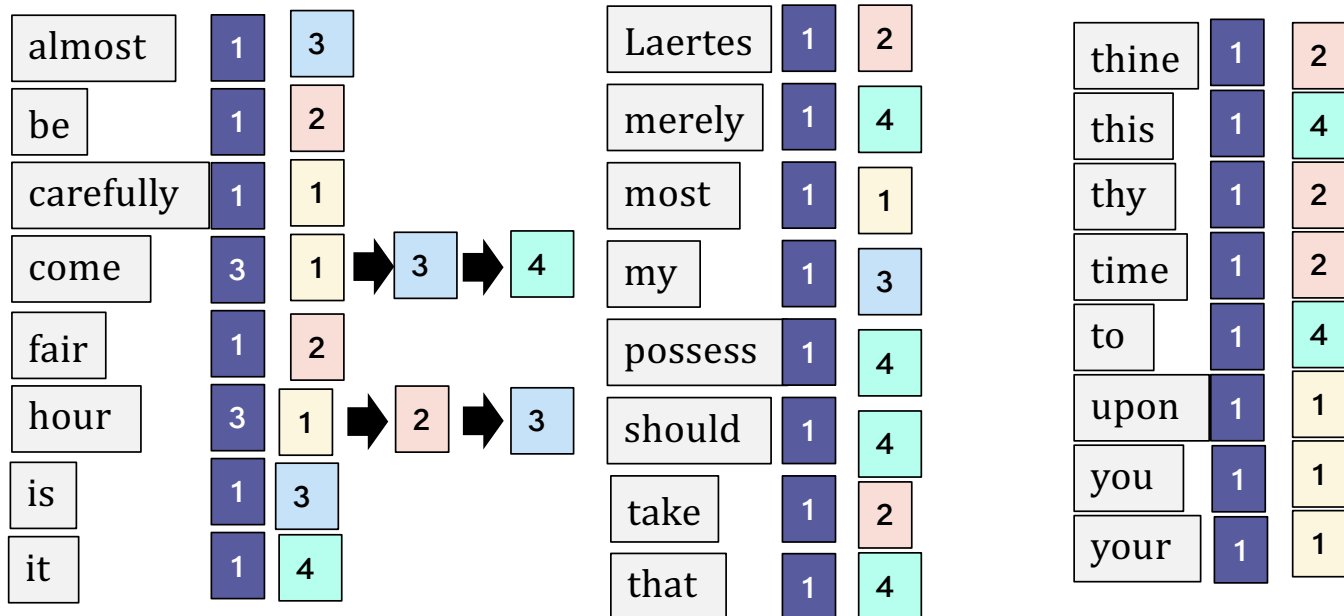
Lemmatization can help with language
that have **more morphology**



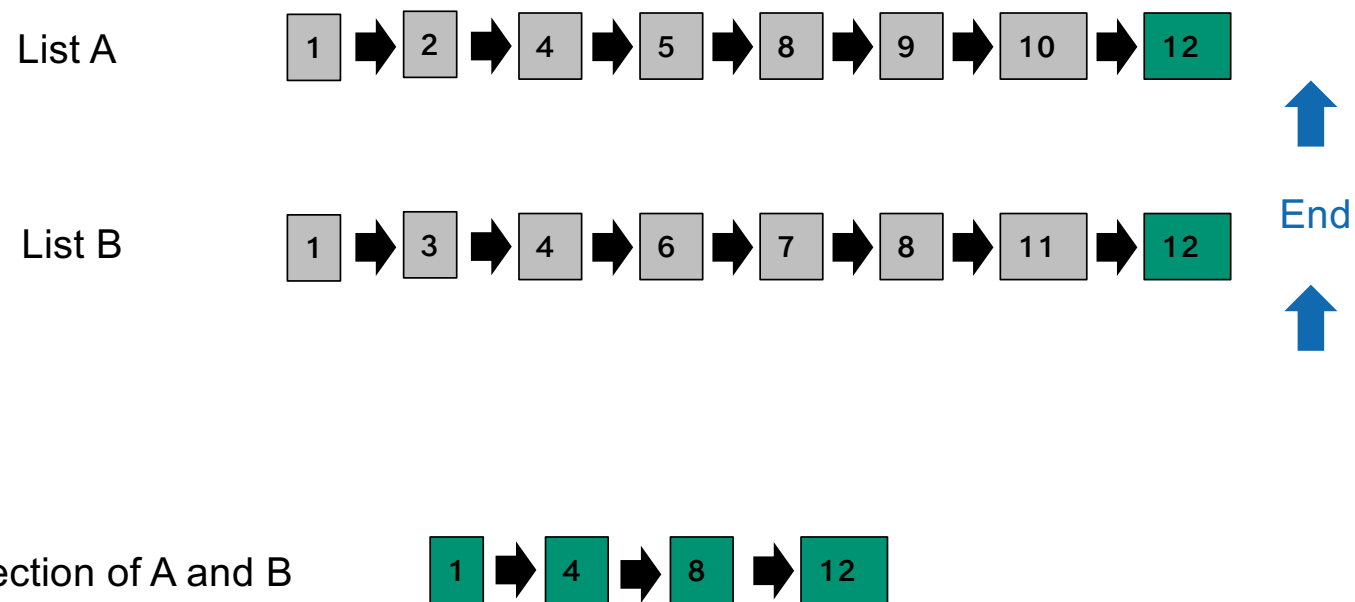


Skip lists

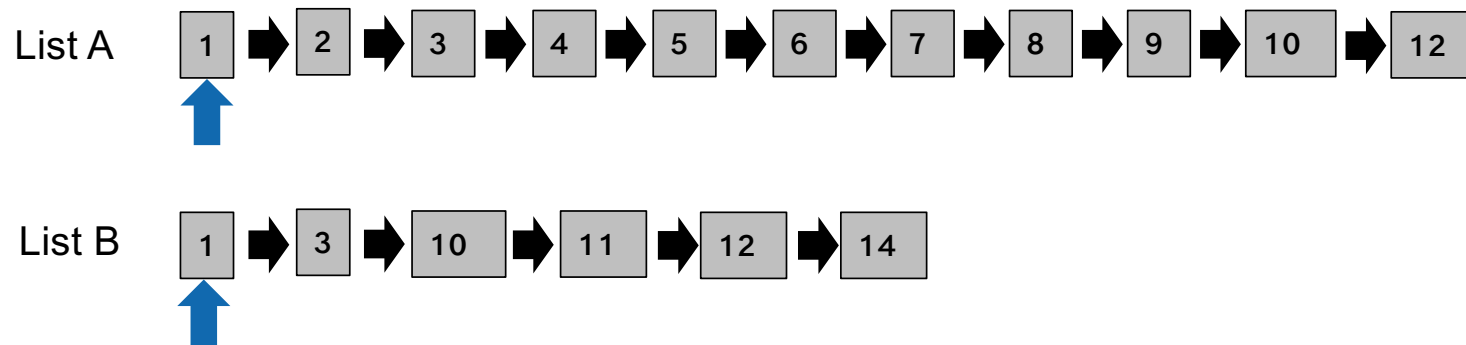
Reminder: Postings (Standard Inverted Index)



Reminder: Intersection algorithm

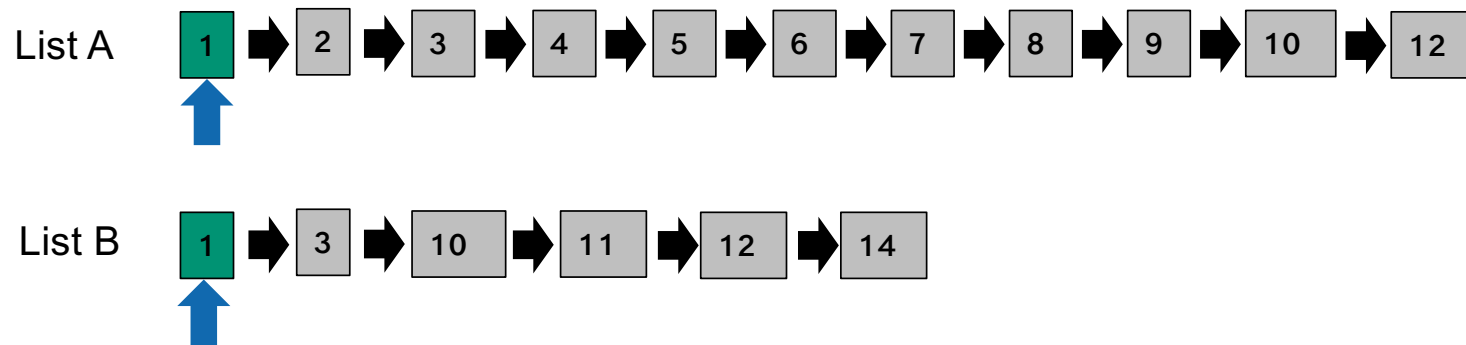


Another example



Intersection of A and B

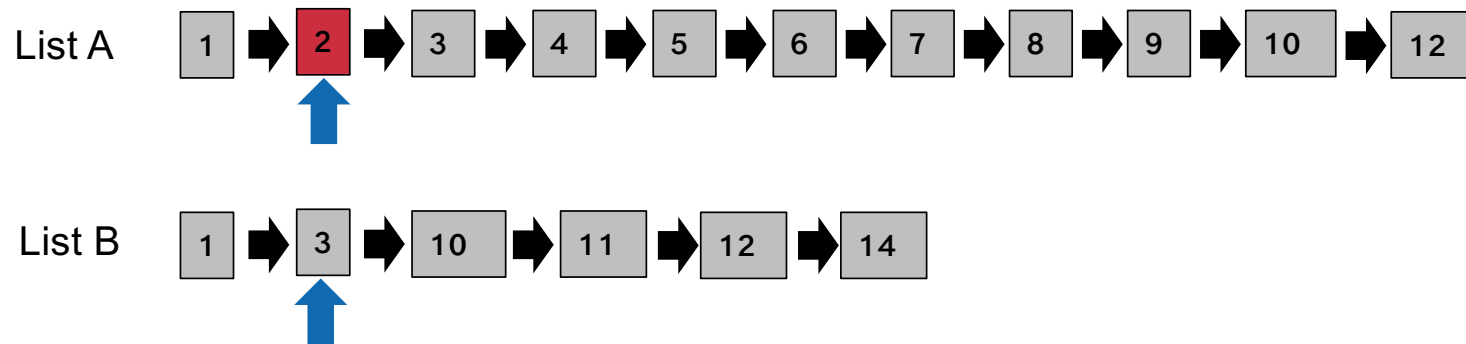
Another example



Intersection of A and B

The intersection of List A and List B is the value 1, which is highlighted in green.

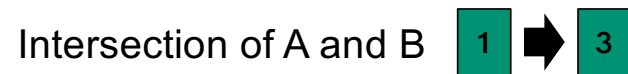
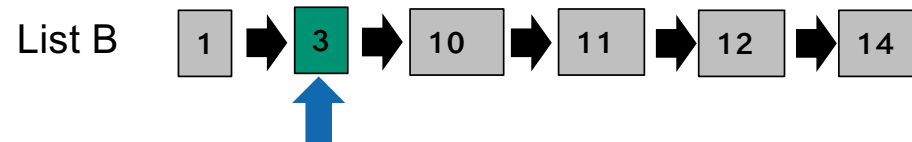
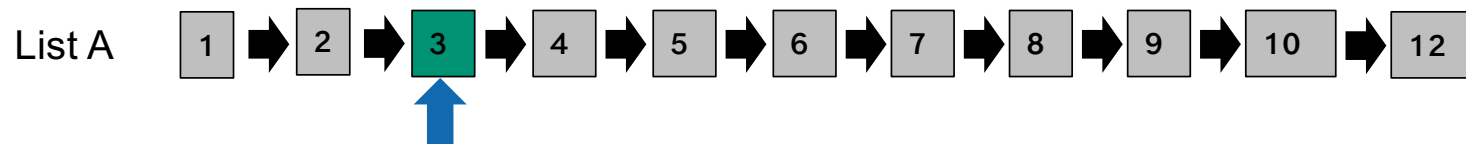
Another example



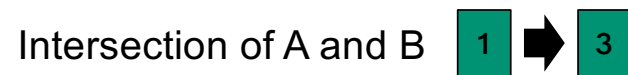
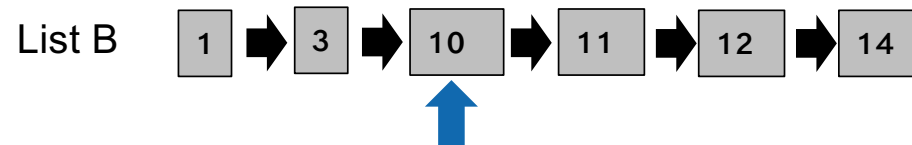
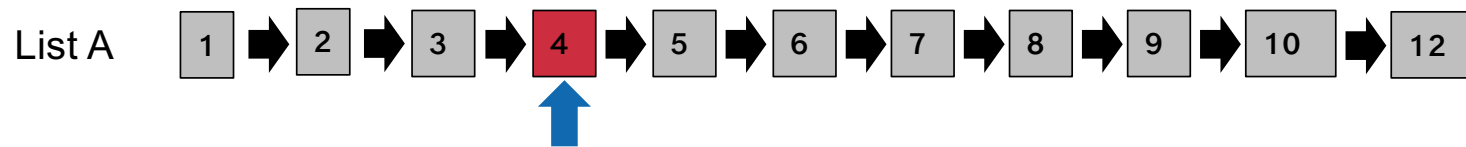
Intersection of A and B

1

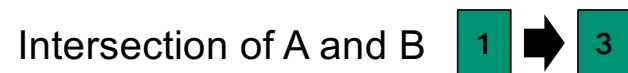
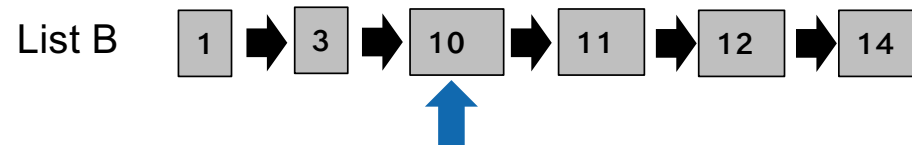
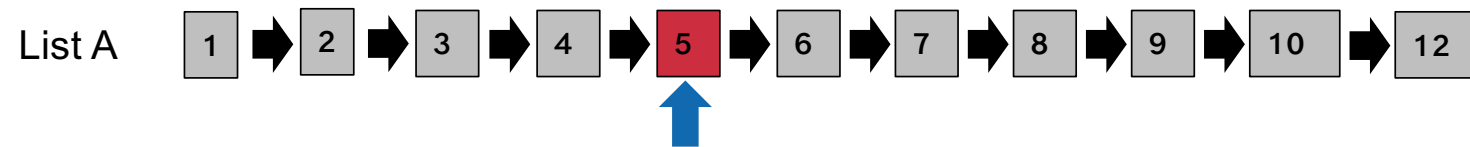
Another example



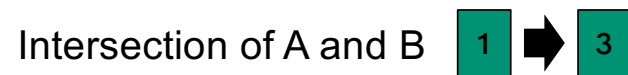
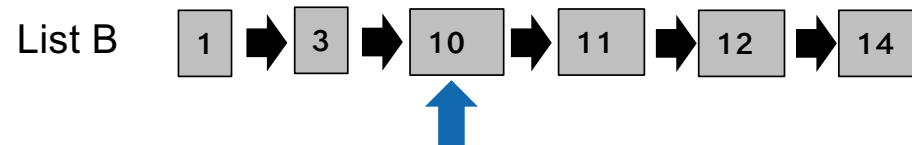
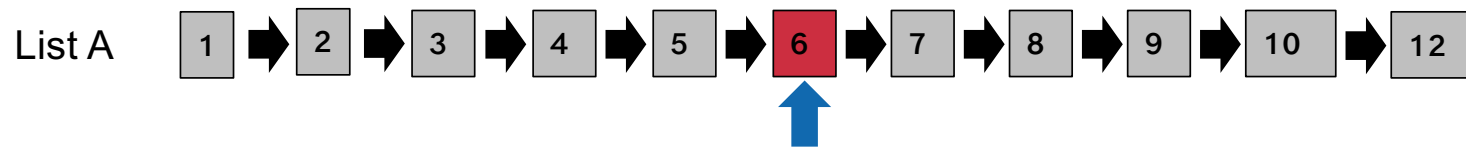
Another example



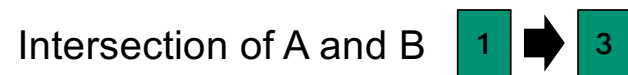
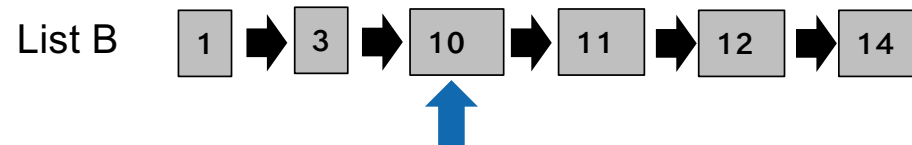
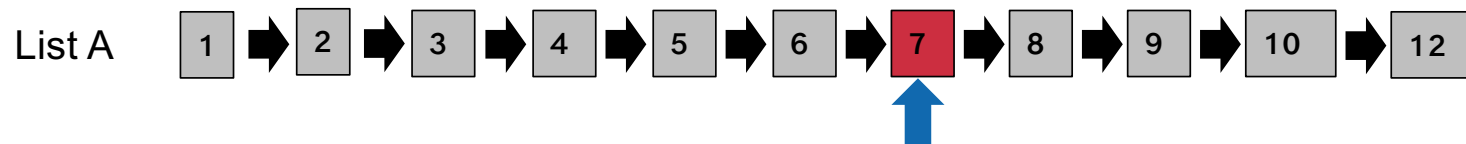
Another example



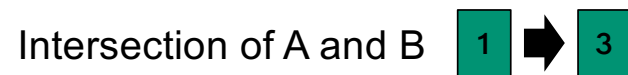
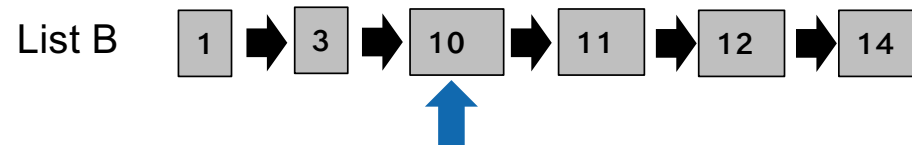
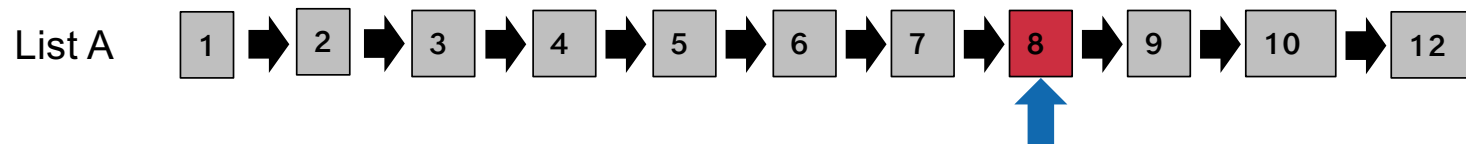
Another example



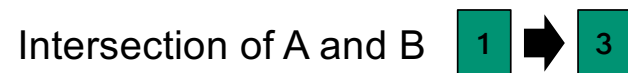
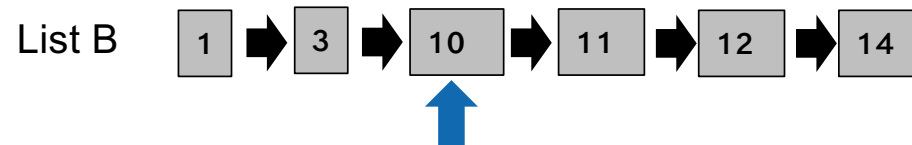
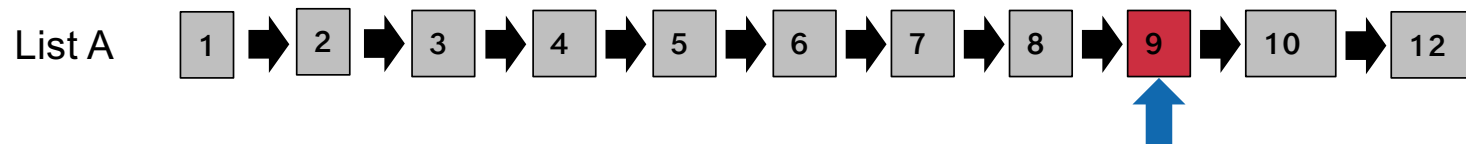
Another example



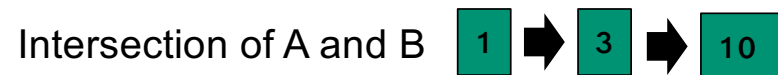
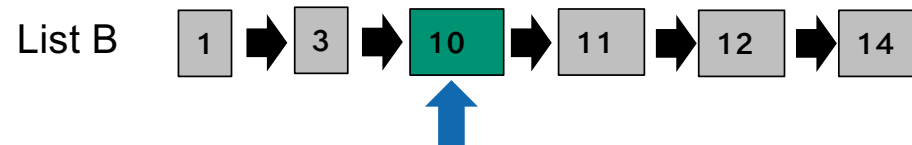
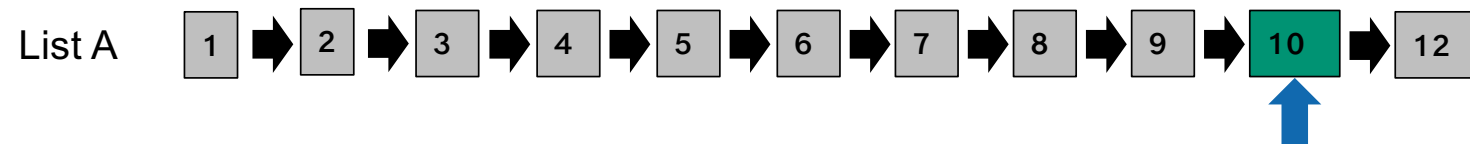
Another example



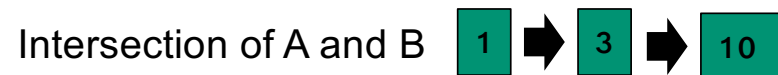
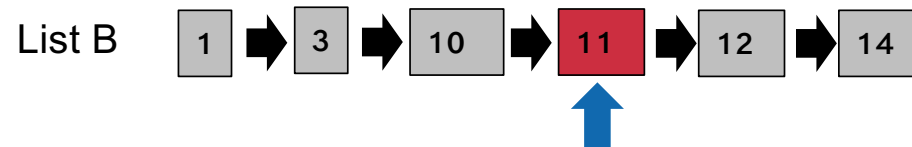
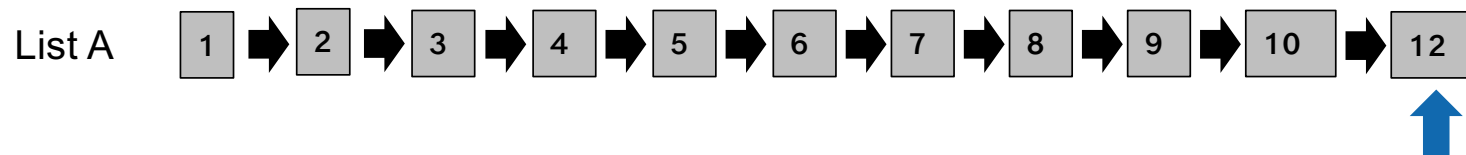
Another example



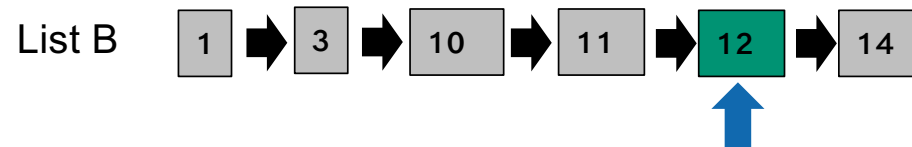
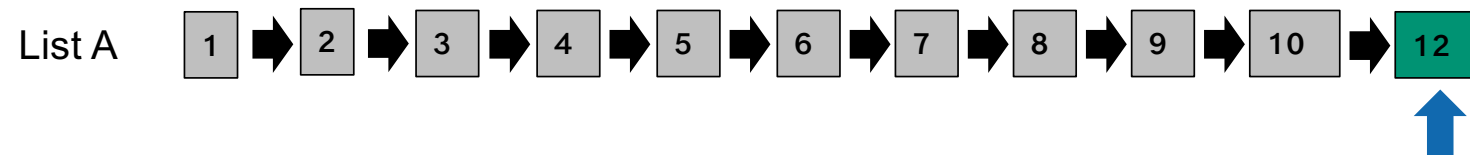
Another example



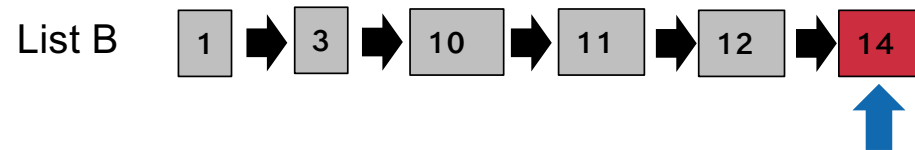
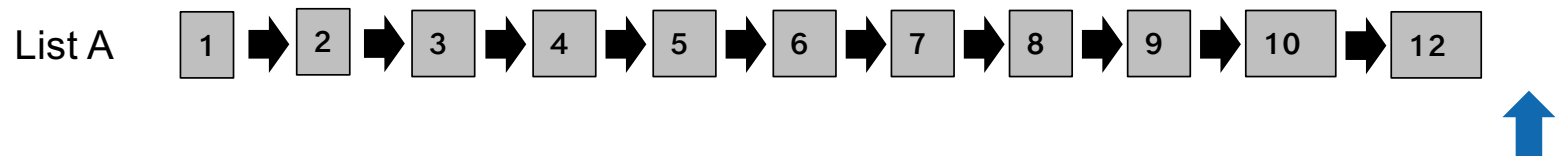
Another example



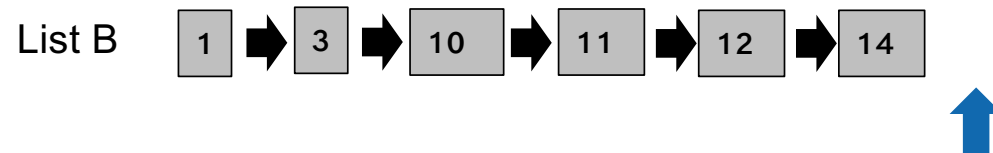
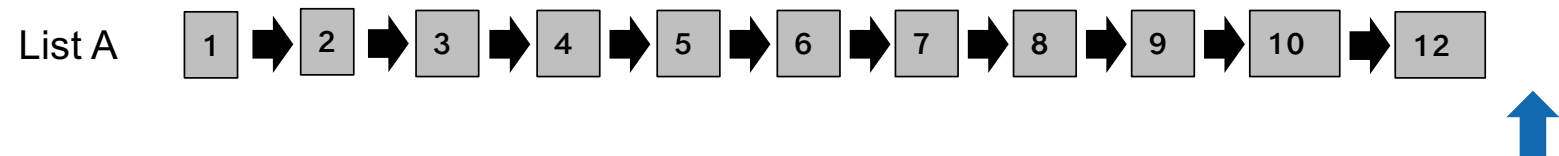
Another example



Another example



Another example



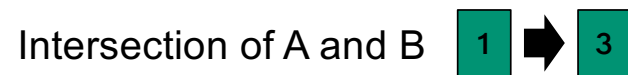
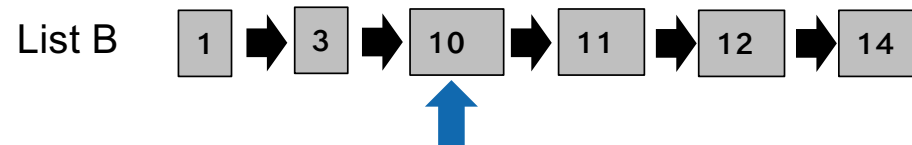
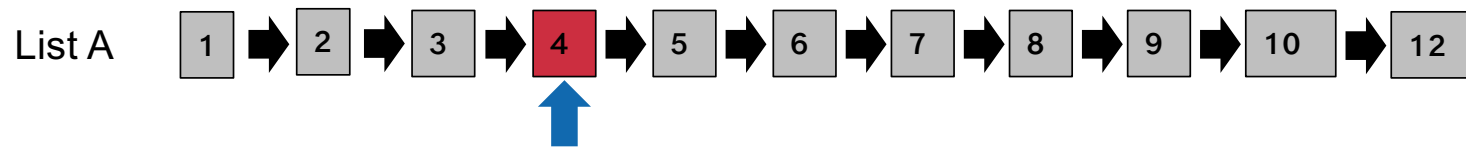
Another example



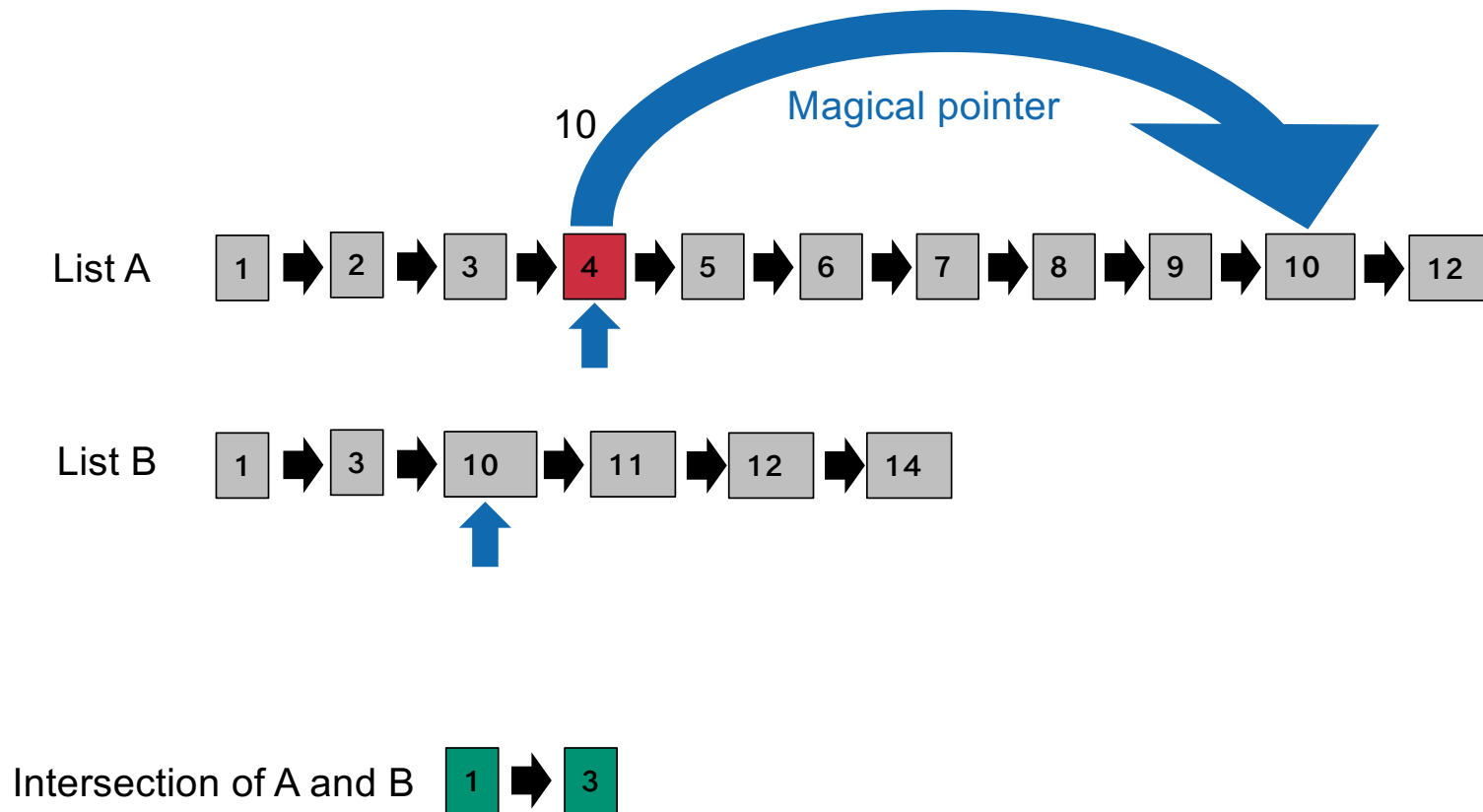
How could we make this

faster?

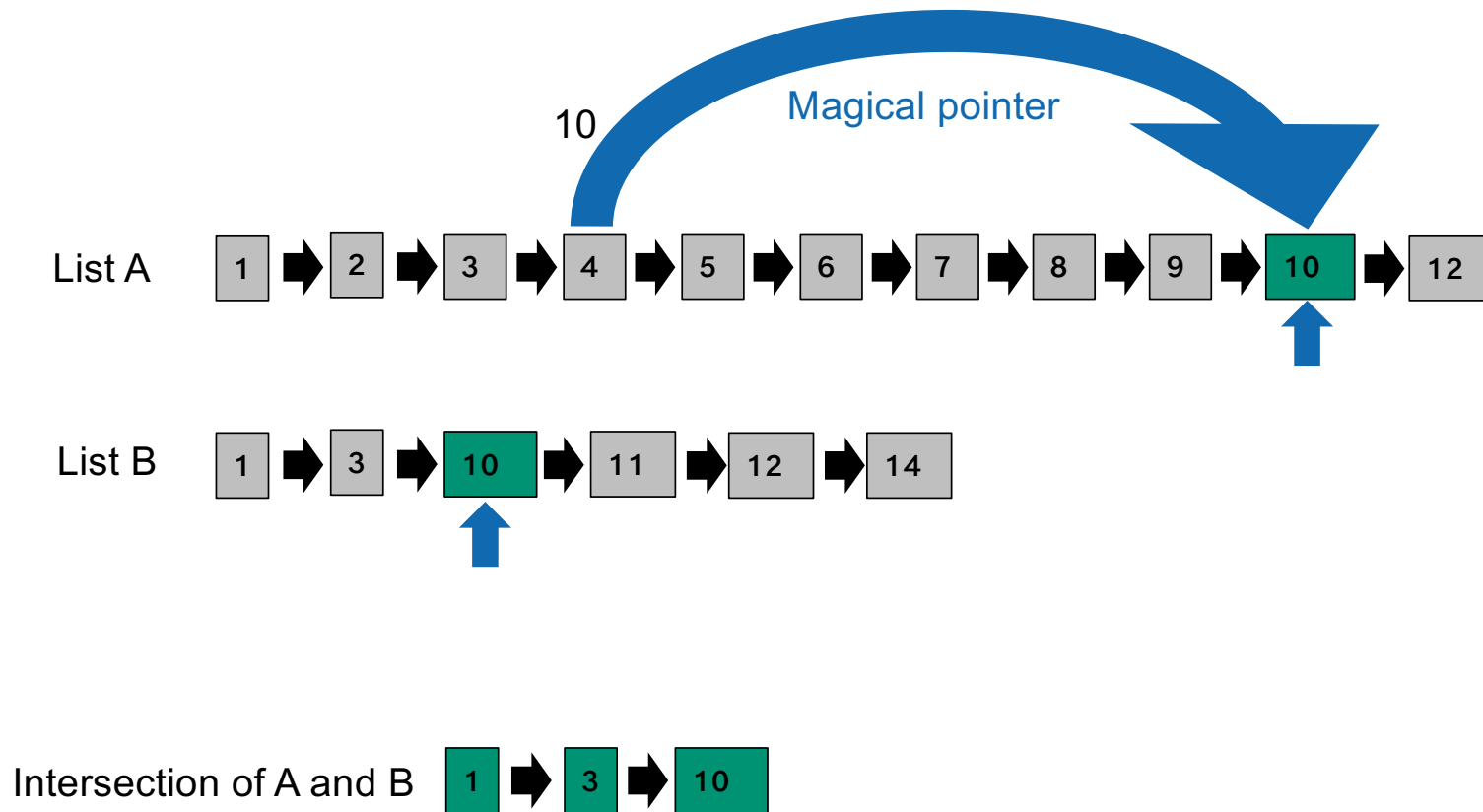
Another example



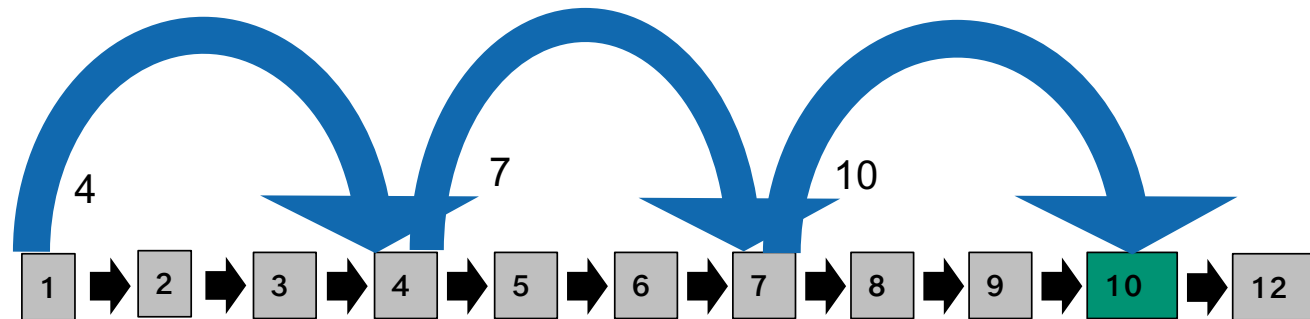
Another example



Another example



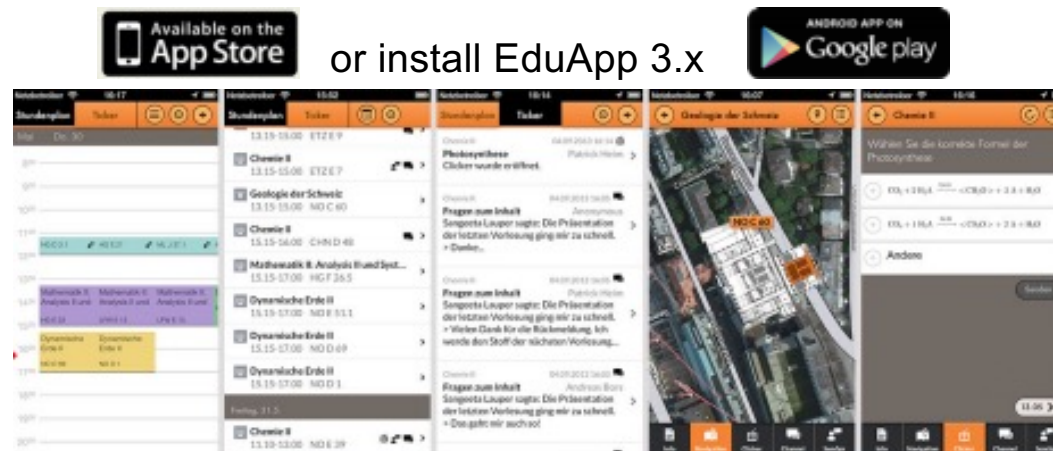
In practice



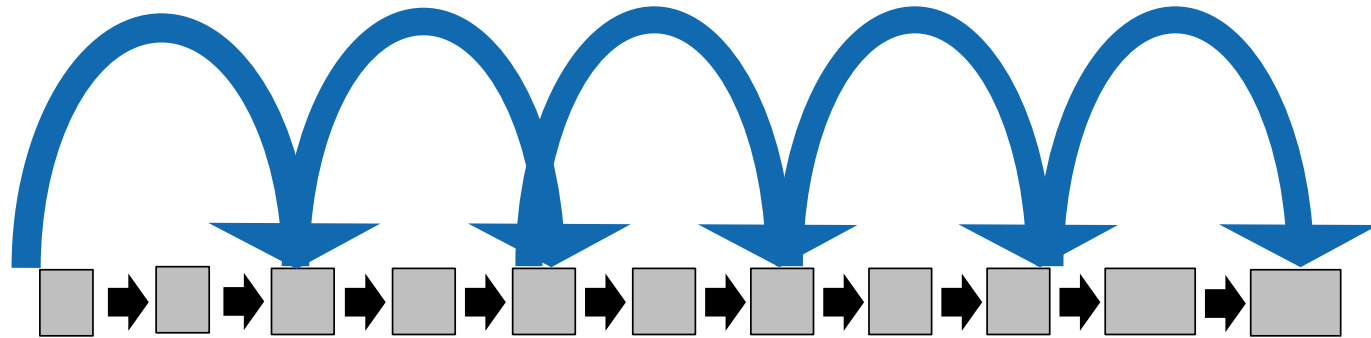
Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>

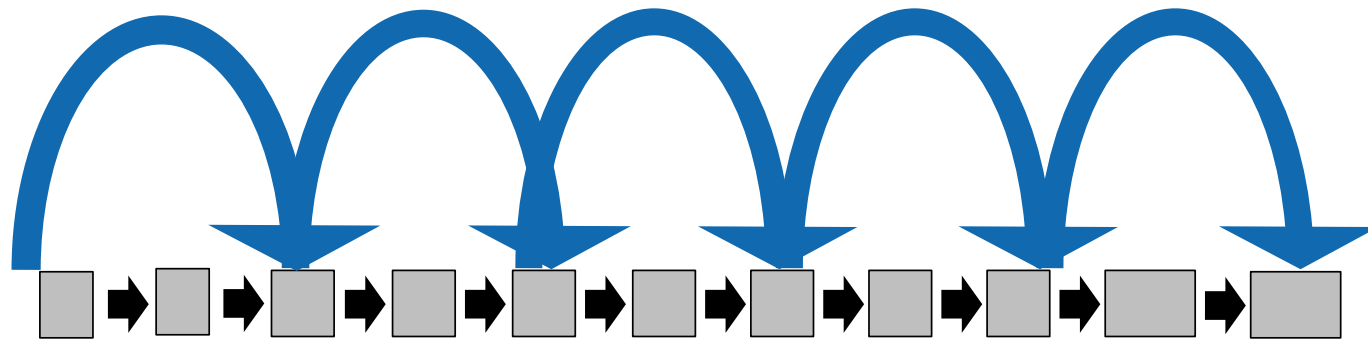


In practice



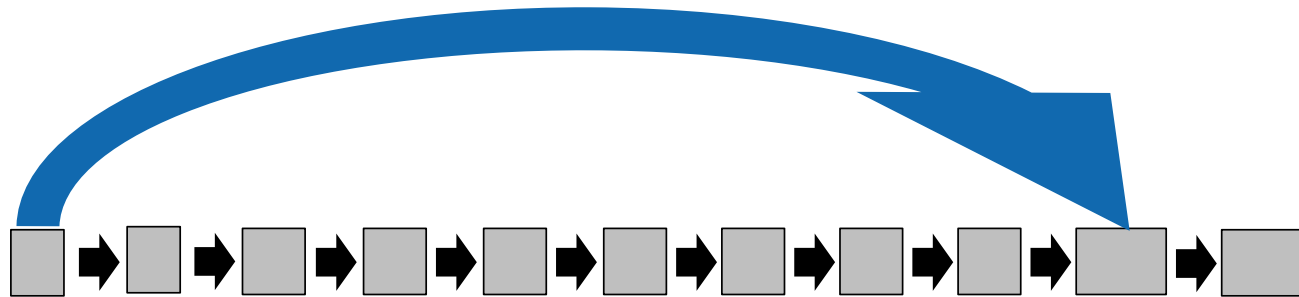
Two short skips:

In practice



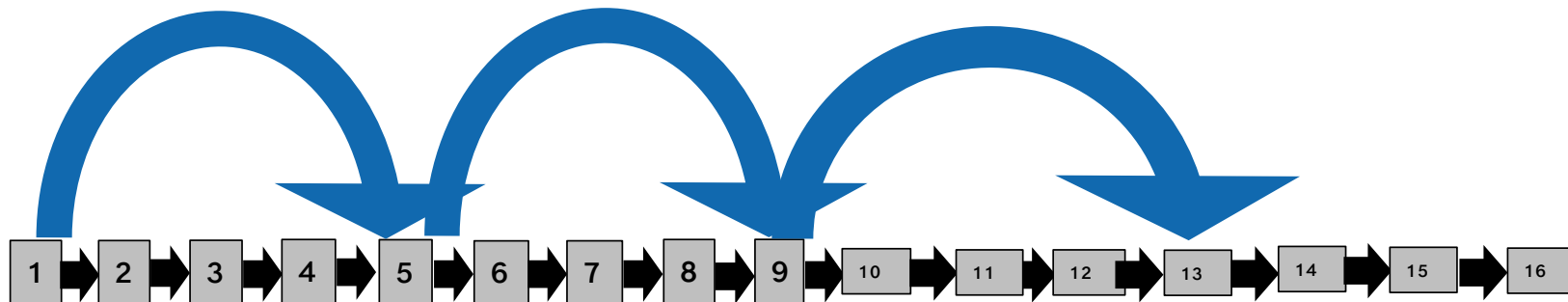
Too short skips:
many comparisons
waste of space

In practice



Too long skips:
few comparisons
not many real opportunities to skip

In practice



In practice $\sqrt{\text{Number of postings}}$



Phrase search

Phrase queries

Google

eth zürich

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About 16,000,000 results (0.44 seconds)

ETH Zürich - Homepage | ETH Zürich
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Logo der ETH Zürich, Suche, de, en, Menü, Startseite · News & Veranstaltungen · Zur Startseite;
Zurück; News & Veranstaltungen · ETH-News · Zur Startseite · News & Veranstaltungen · ETH-News ·
Archiv · Redaktion · Newsletter und Abo · Zur Startseite · ETH-News · Newsletter und Abo · Abmelden/
ändern · Mobile Apps.

Results from ethz.ch

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31.01.2018 | Prof. Ernst Hafen |
Health. Embracing innovation ...

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Computer Science Department at
ETH Zurich. The department ...

Porträt
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Research | ETH Zurich
ETH's research is focused on the
needs of society – be that at ...

(ETH) Zurich Physics ...
Institutes and Groups · Research
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Hours ▾

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B ETH Zürich - Clausiusbar
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C Graphische Sammlung ETH Zürich
ETH, Rämistrasse 101 · 044 632 40 46
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eth zürich

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About 3.310.000 results (0,47 seconds)

ETH Zürich - Homepage | ETH Zürich

www.ethz.ch/

Die ETH Zürich unterstützen. Tragen Sie dazu bei, dass die ETH Zürich auch in Zukunft flexibel auf Herausforderungen reagieren und neue Akzente setzen kann. Mehr Informationen für Gönner und Partner ...

Results from ethz.ch

ETH Zurich

31.01.2018 | Prof. Ernst Hafen | Health. Embracing innovation ...

ETH Zurich Physics ...

Institutes and Groups - Research Overview - People - BSc Physics

Porträt

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Department of Architecture

Prof. Dr. Philippe Block has been named this year's recipient of ...

ETH - Computer Science

Computer Science Department at ETH Zurich. The department ...

Schweizerisches Nationalmuseum

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ETH, Rämistrasse 101 · 044 632 40 46

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More locations

ETH Zurich - Wikipedia

With a posting list

"ETH Zürich"

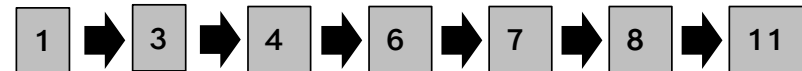
Quotes = phrase search

With a posting list

ETH



Zurich

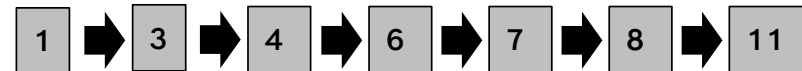


With a posting list

ETH



Zurich



We have no information on the proximity of the terms!

Phrase search: approaches

Phrase search: approaches

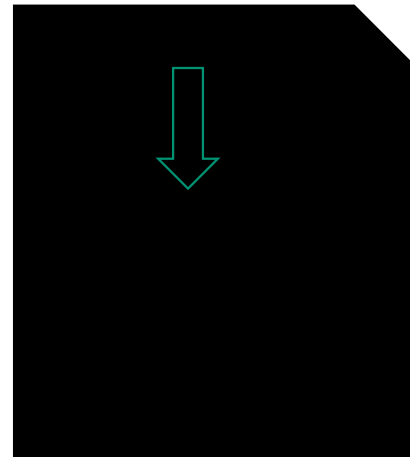


Biword indices

Phrase search: approaches



Biword indices



Positional indices

Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.

Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



Index

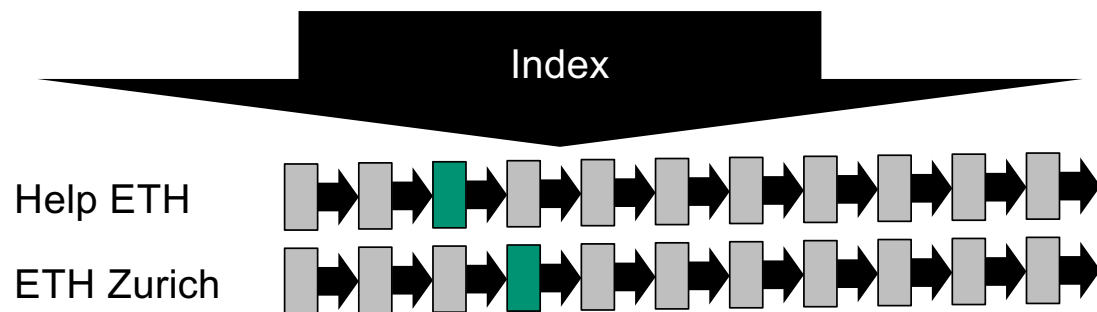
Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



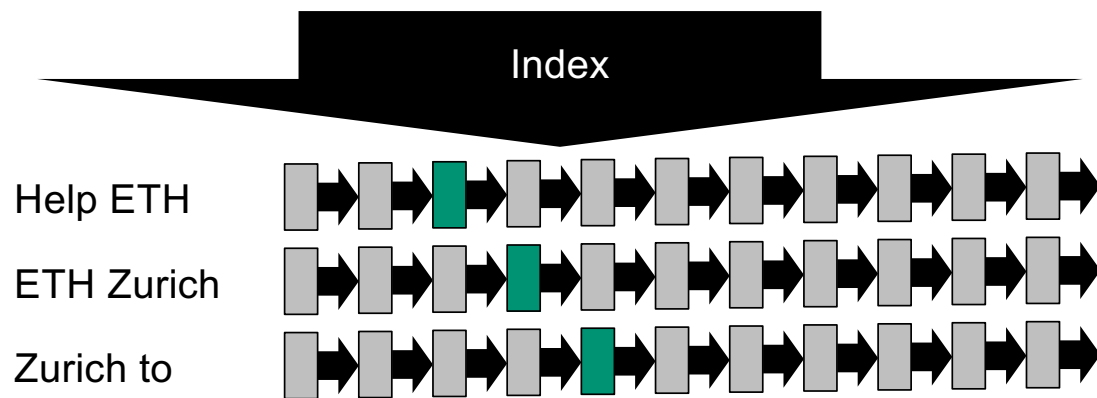
Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



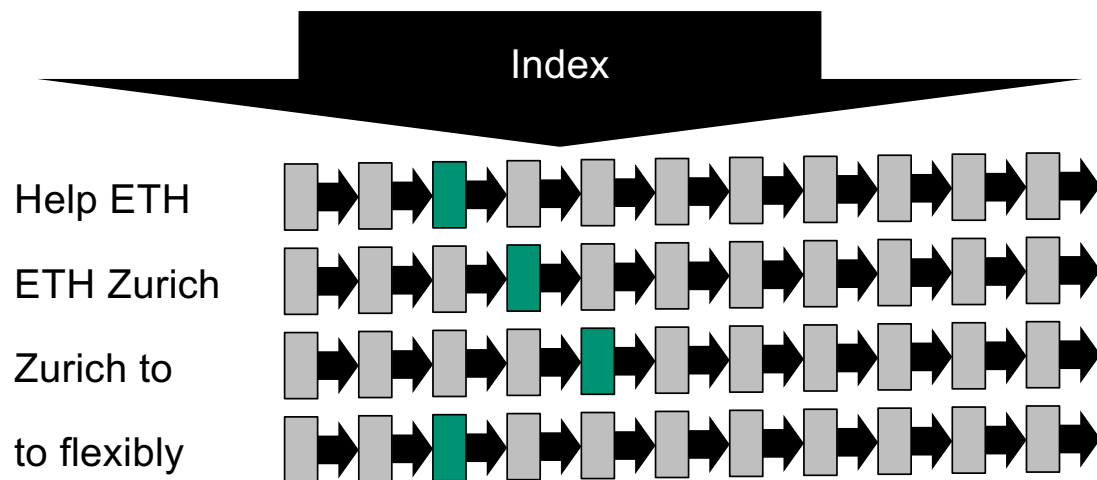
Bi-word indices

Help ETH Zurich flexibly react to new challenges and to set new accents in the future.



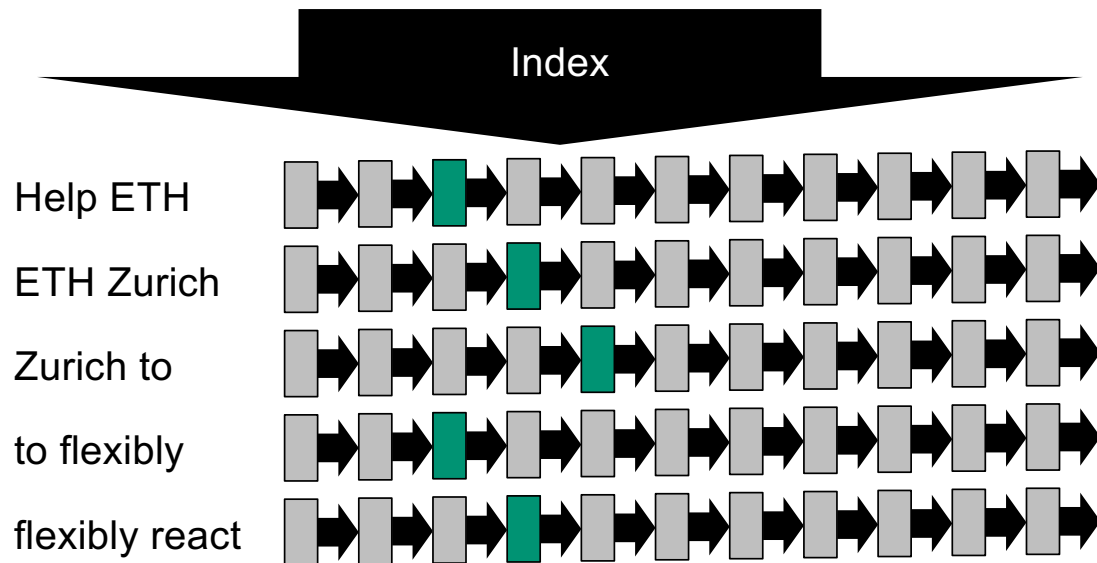
Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



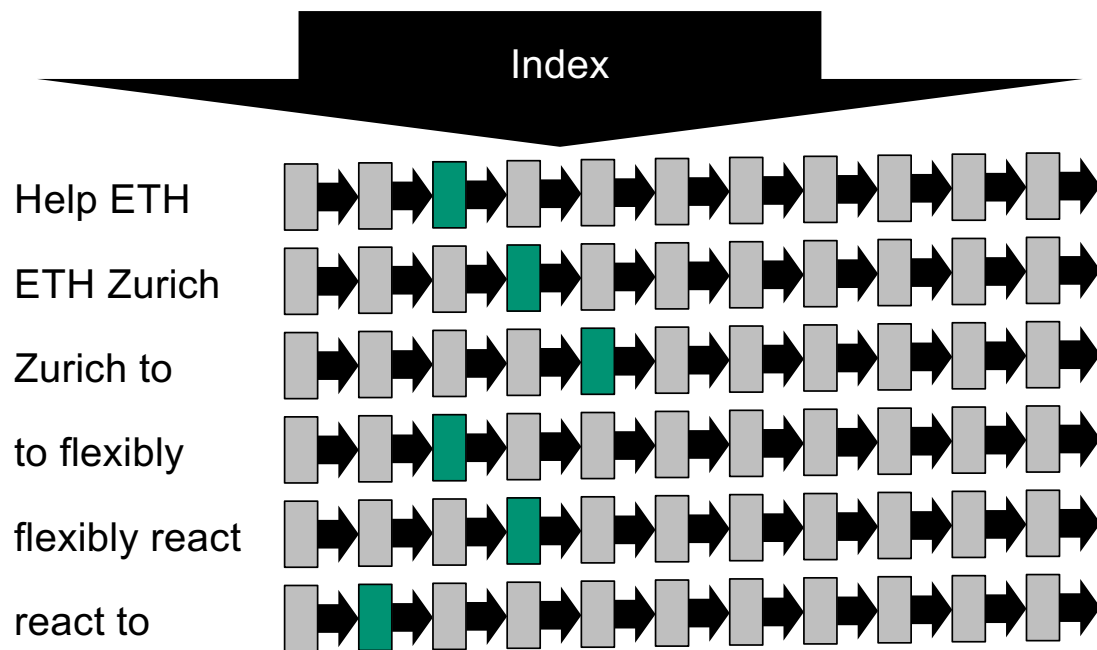
Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



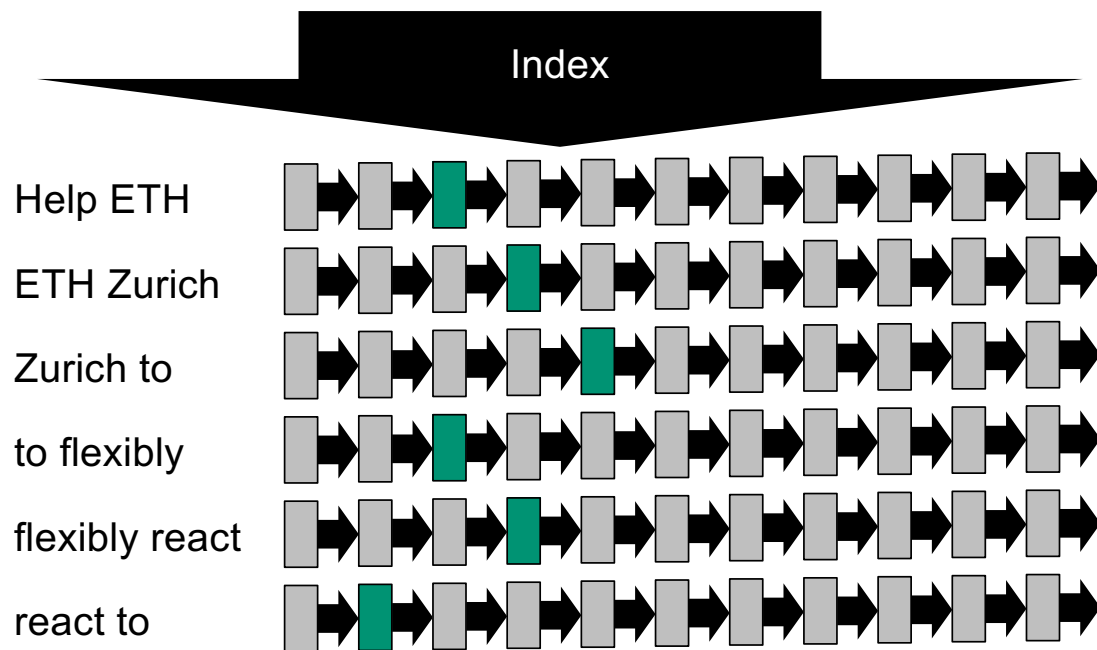
Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



Bi-word indices

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



Bi-word indices

"ETH Zurich"

Query

Help ETH

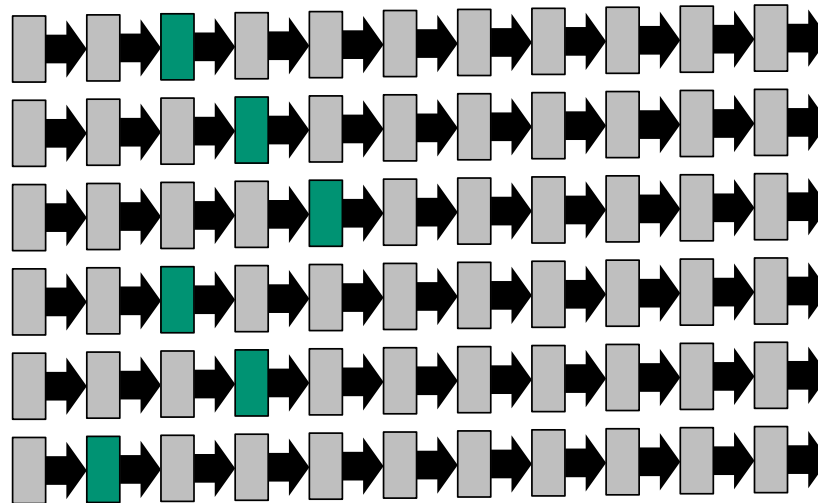
ETH Zurich

Zurich to

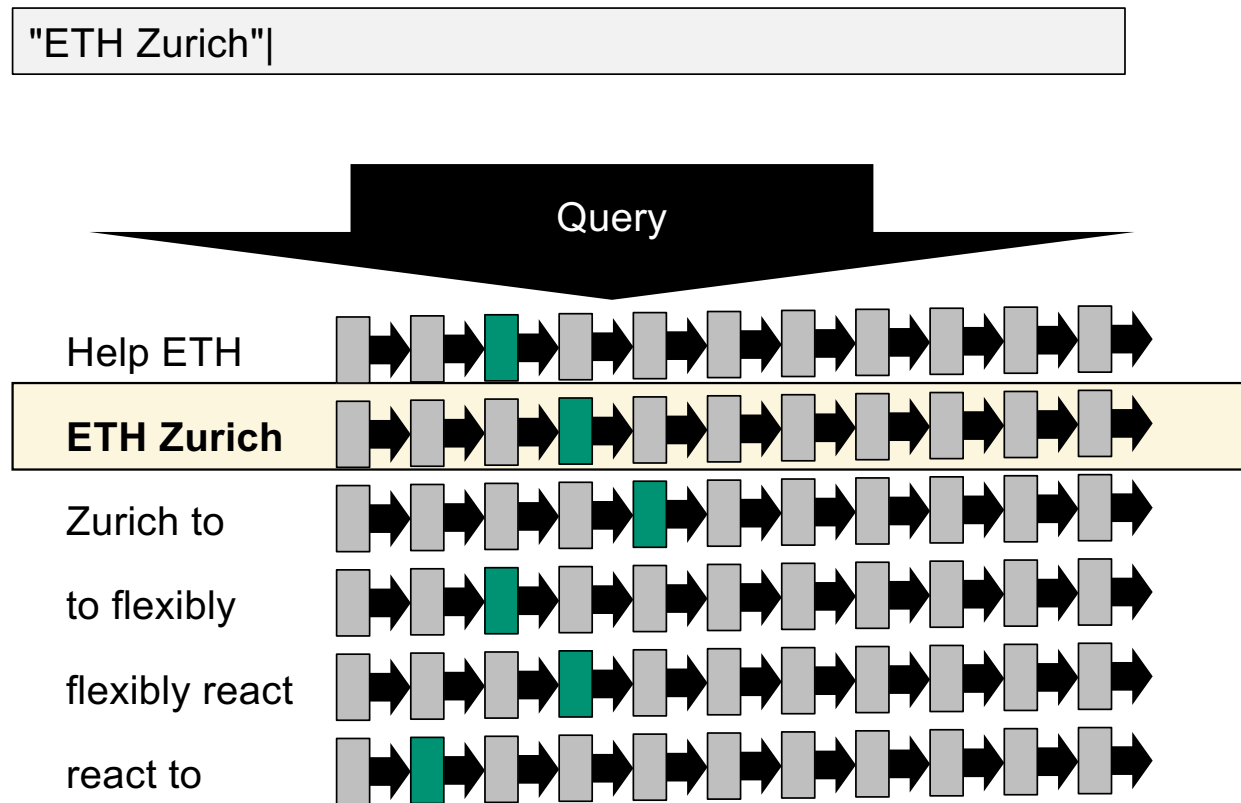
to flexibly

flexibly react

react to



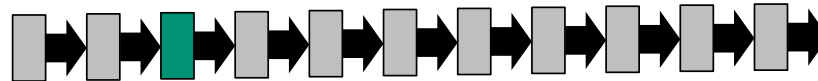
Bi-word indices



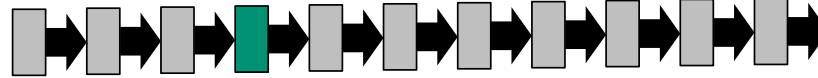
Bi-word indices

"Help ETH Zurich to flexibly react"

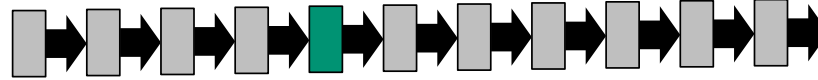
Help ETH



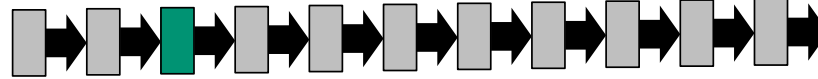
ETH Zurich



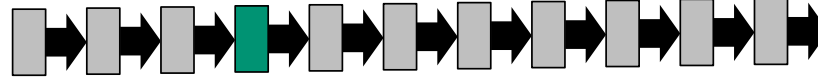
Zurich to



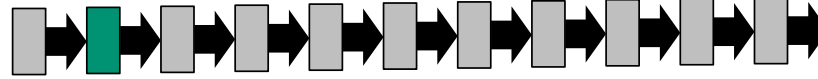
to flexibly



flexibly react



react to

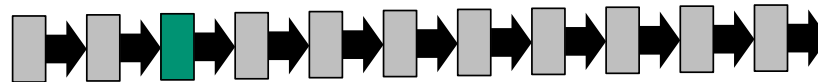


Bi-word indices

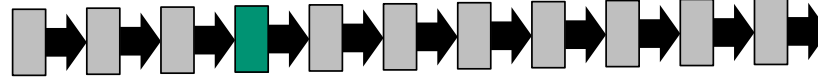
"Help ETH Zurich to flexibly react"

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"

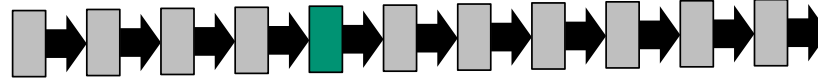
Help ETH



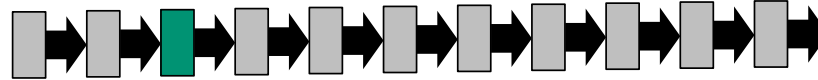
ETH Zurich



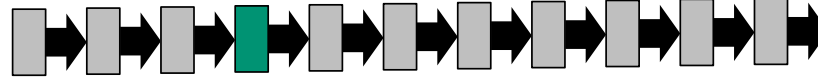
Zurich to



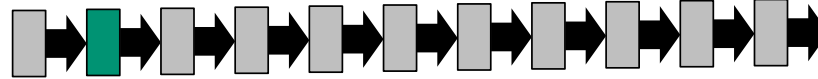
to flexibly



flexibly react



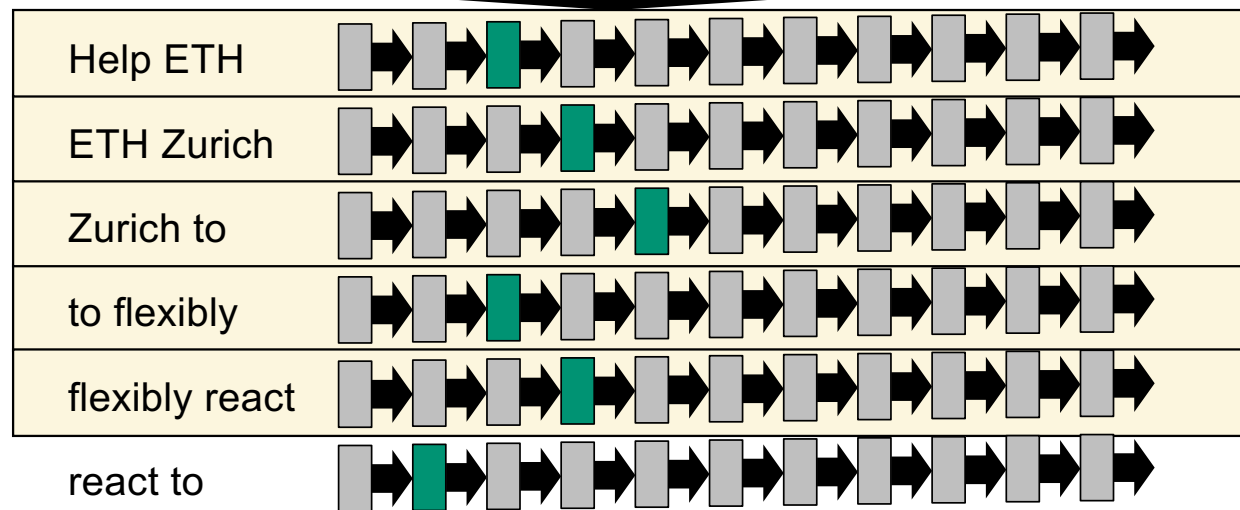
react to



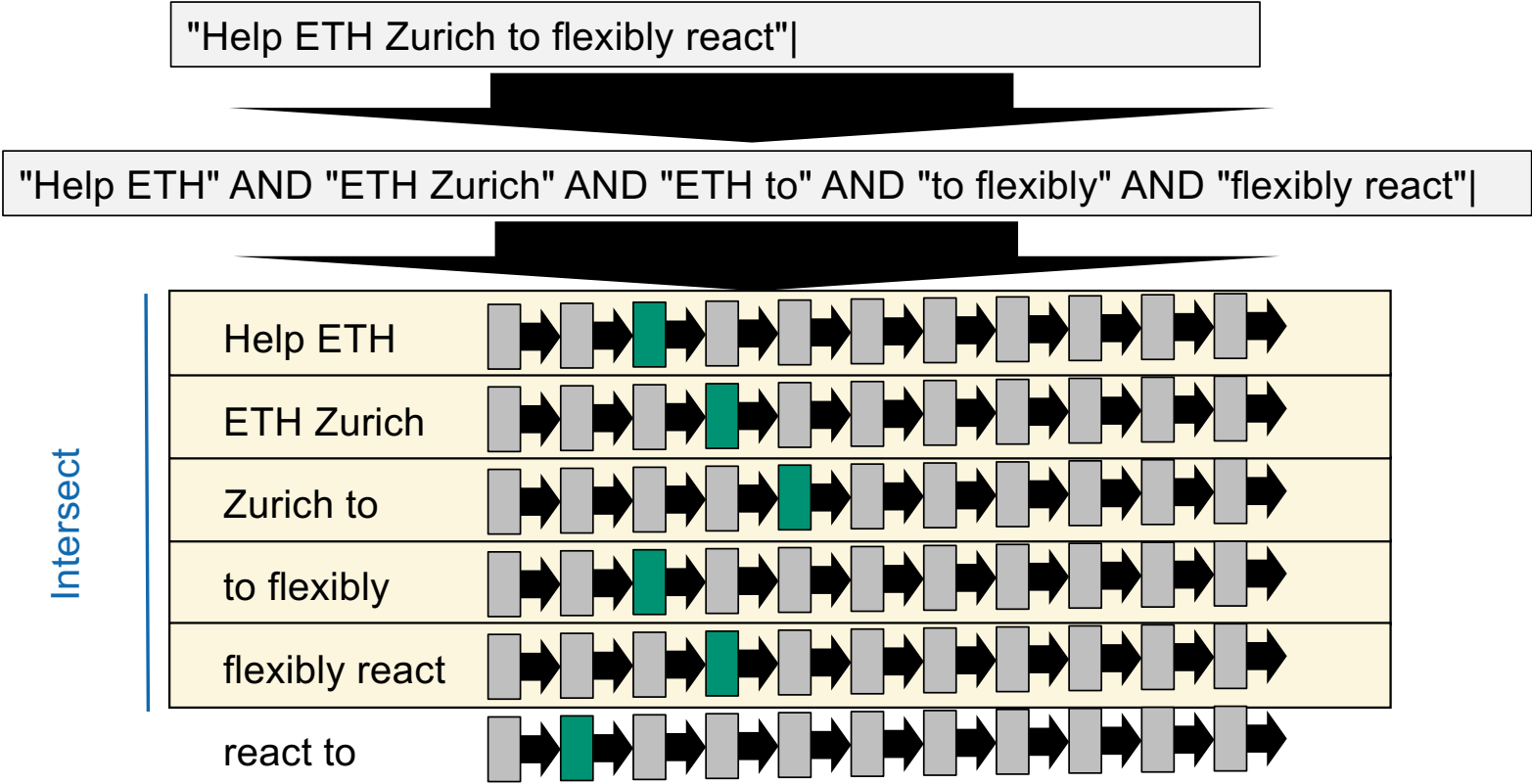
Bi-word indices

"Help ETH Zurich to flexibly react"

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"



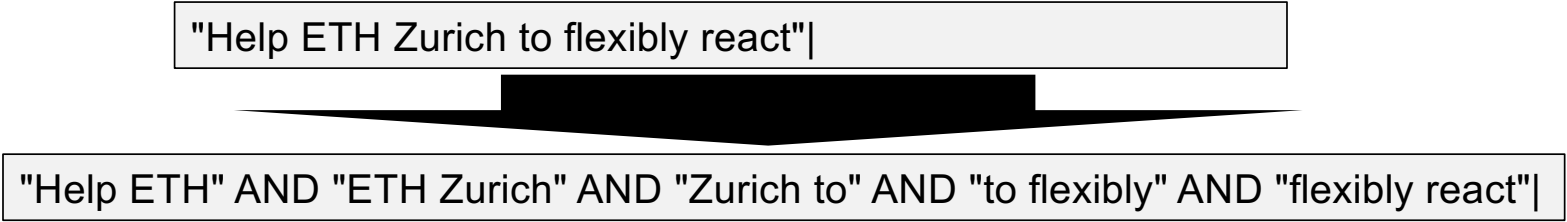
Bi-word indices



Inconvenient?

Inconvenient?

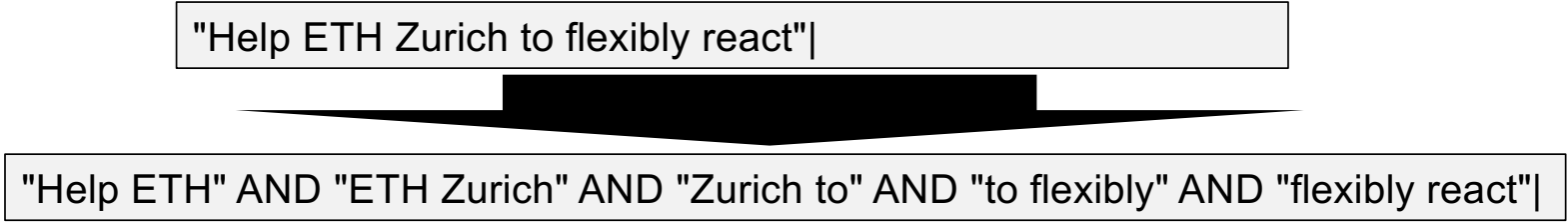
"Help ETH Zurich to flexibly react"|



"Help ETH" AND "ETH Zurich" AND "Zurich to" AND "to flexibly" AND "flexibly react"|

Inconvenient?

"Help ETH Zurich to flexibly react"|



"Help ETH" AND "ETH Zurich" AND "Zurich to" AND "to flexibly" AND "flexibly react"|

Help ETH Zurich to introduce techniques to flexibly react...

Inconvenient?

"Help ETH Zurich to flexibly react"|

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"|

Help ETH Zurich to introduce techniques to flexibly react...



Inconvenient?

"Help ETH Zurich to flexibly react"|

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"|

Help ETH Zurich to introduce techniques to flexibly react...



Inconvenient?

"Help ETH Zurich to flexibly react"|

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"|

Help ETH Zurich to introduce techniques to flexibly react...



Inconvenient?

"Help ETH Zurich to flexibly react"|

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"|

Help ETH Zurich to introduce techniques to flexibly react...



Inconvenient?

"Help ETH Zurich to flexibly react"|

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"|

Help ETH Zurich to introduce techniques to flexibly react...



Inconvenient?

"Help ETH Zurich to flexibly react"|

"Help ETH" AND "ETH Zurich" AND "ETH to" AND "to flexibly" AND "flexibly react"|

Help ETH Zurich to introduce techniques to flexibly react...



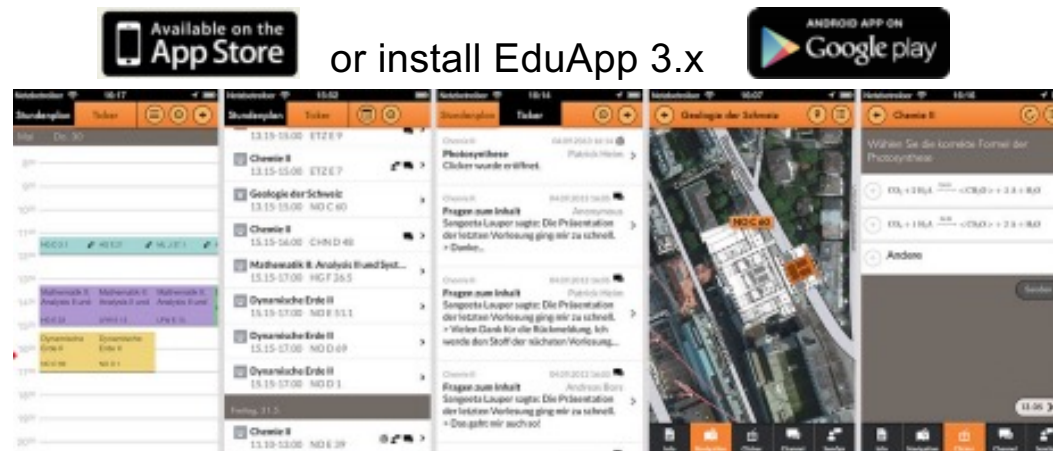
False positive!



Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>



Part-of-speech tagging

Help ETH Zurich to introduce techniques so as to flexibly react...

Part-of-speech tagging

Help ETH Zurich to introduce techniques so as to flexibly react...

Part-of-speech tagging

Help ETH Zurich to introduce techniques so as to flexibly react...

N N N X N N X X X N N

Part-of-speech tagging

NX^*N

Help ETH Zurich to introduce techniques so as to flexibly react...

N N N X N N X X X N N

Part-of-speech tagging

NX^*N

Help ETH Zurich to introduce techniques so as to flexibly react...

N N N X N N X X X N N

Part-of-speech tagging

NX^*N

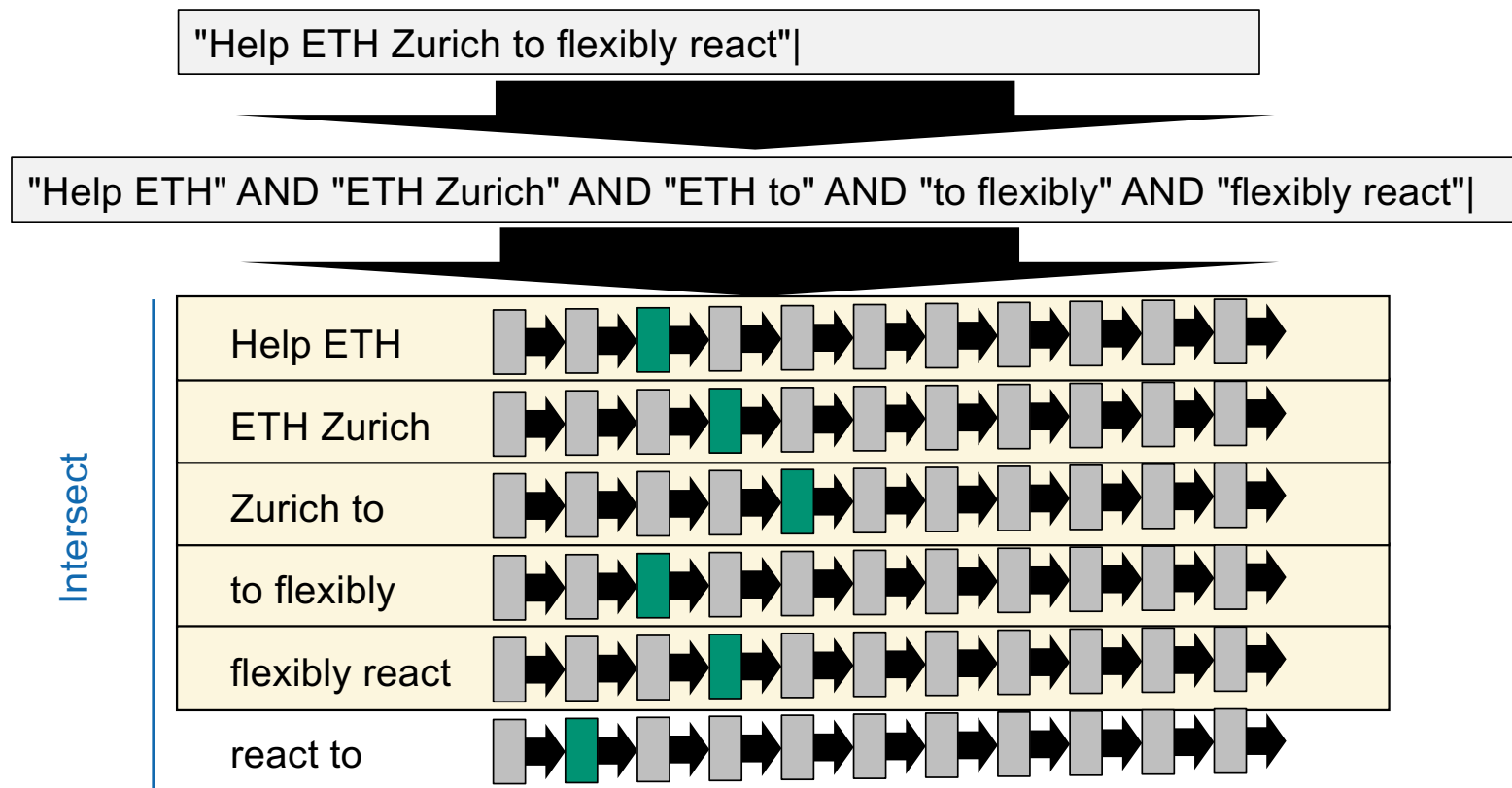
Help ETH Zurich to introduce techniques so as to flexibly react...

N N N X N N X X X N N

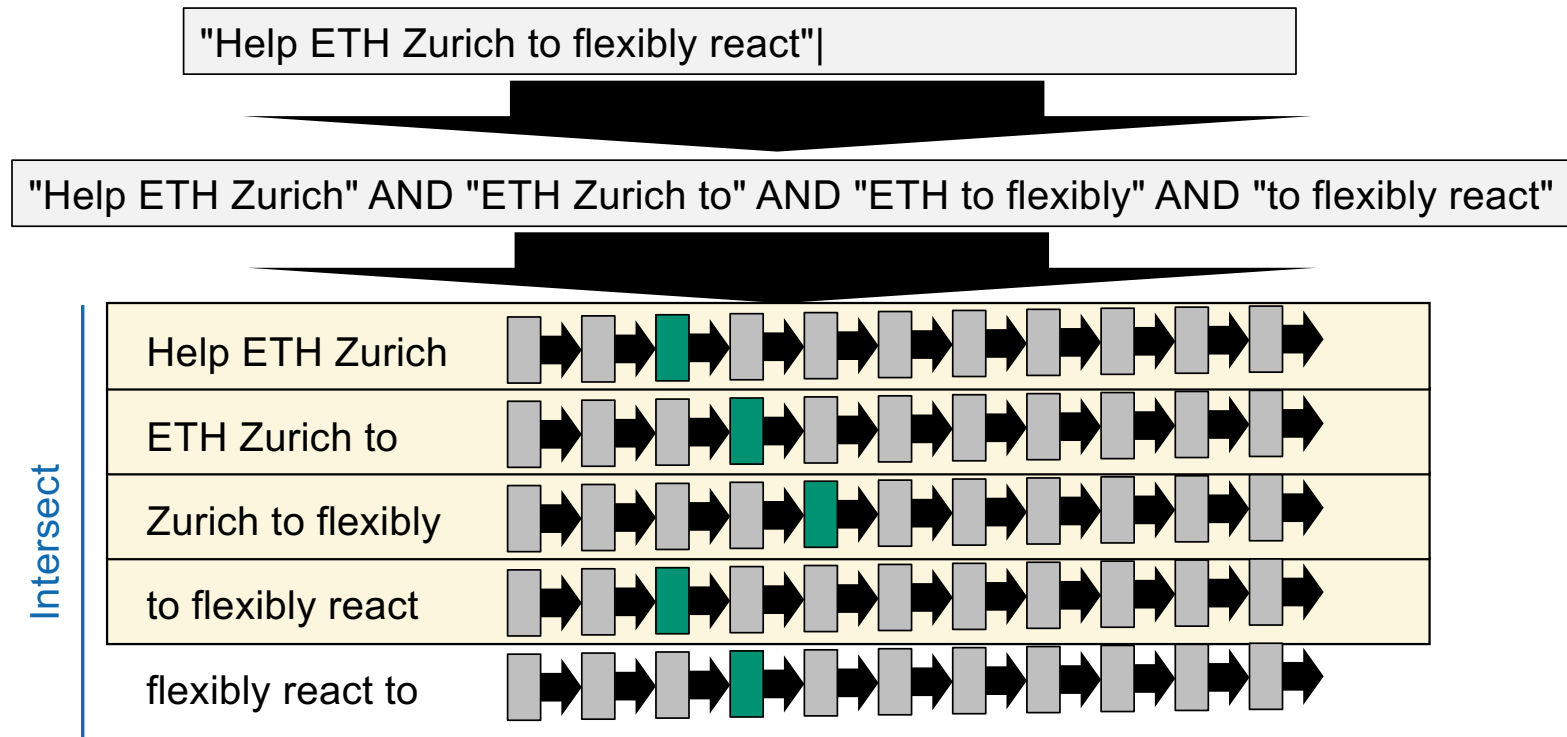


Help ETH
ETH Zurich
Zurich introduce
introduce techniques
techniques flexibly,
flexibly react...

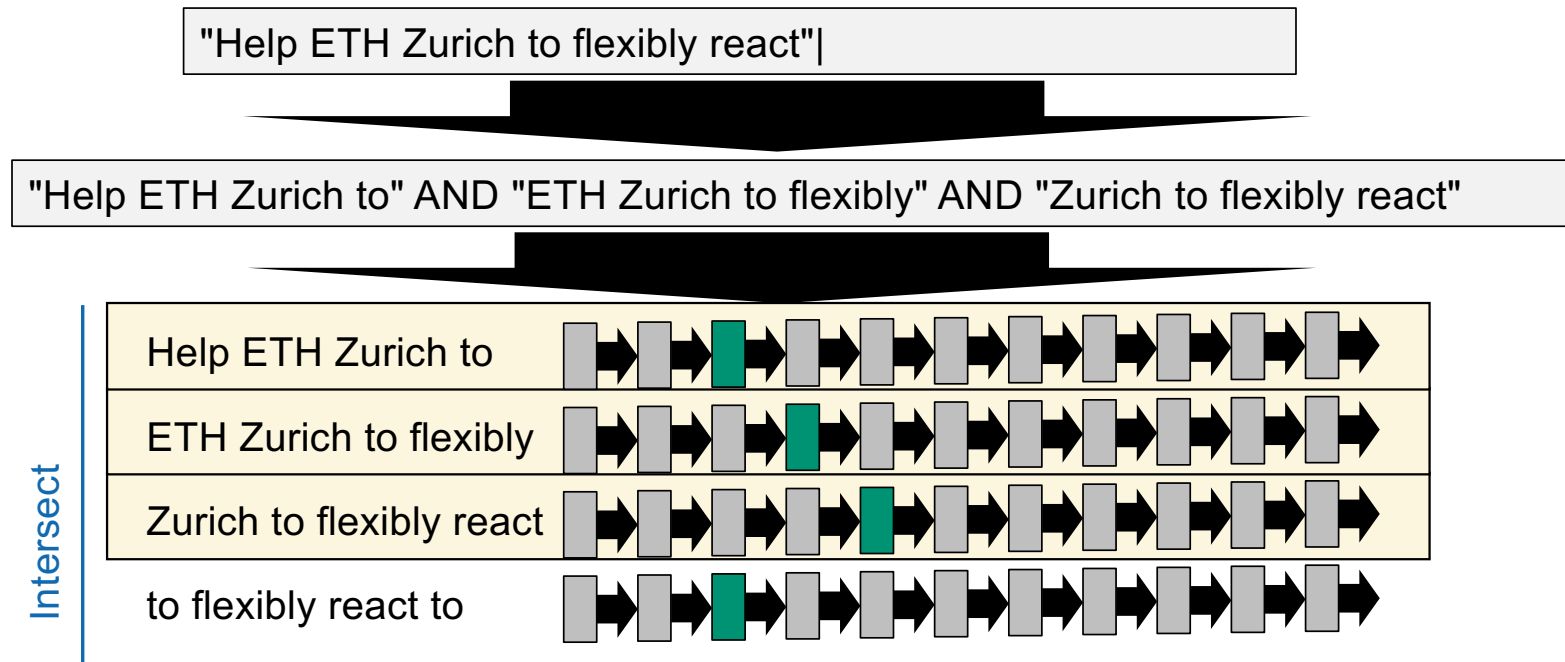
Bi-word indices



Phrase indices (3)



Phrase indices (4)



Improves on false positive issue

"Help ETH Zurich to flexibly react"|

"Help ETH Zurich" AND "ETH Zurich to" AND "Zurich to flexibly" AND "to flexibly react"

Help ETH Zurich to introduce techniques to flexibly react...

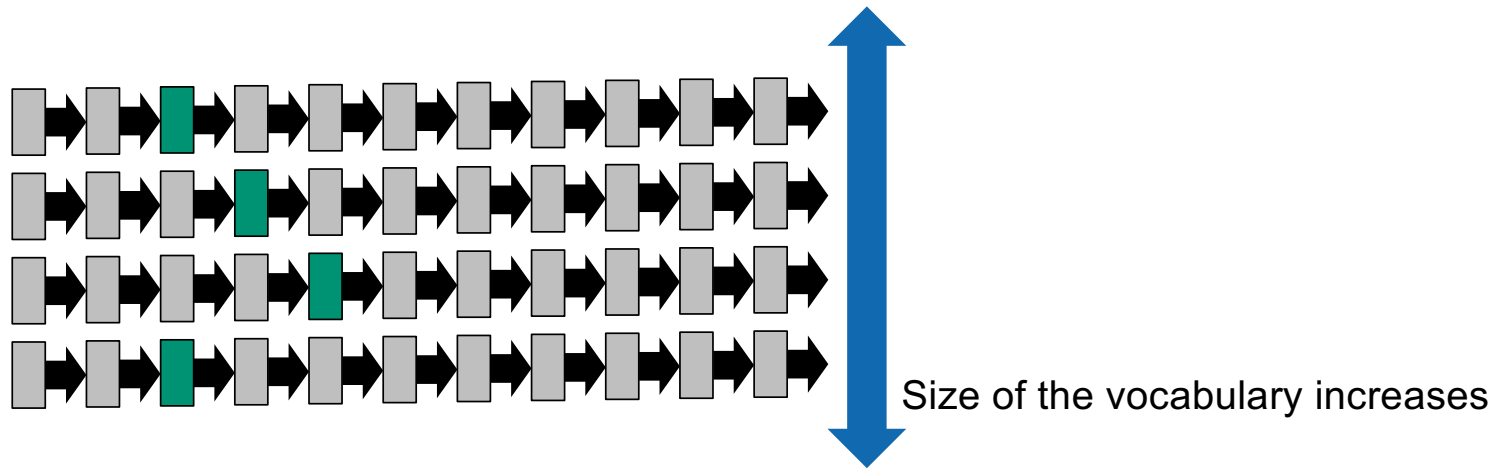


True negative

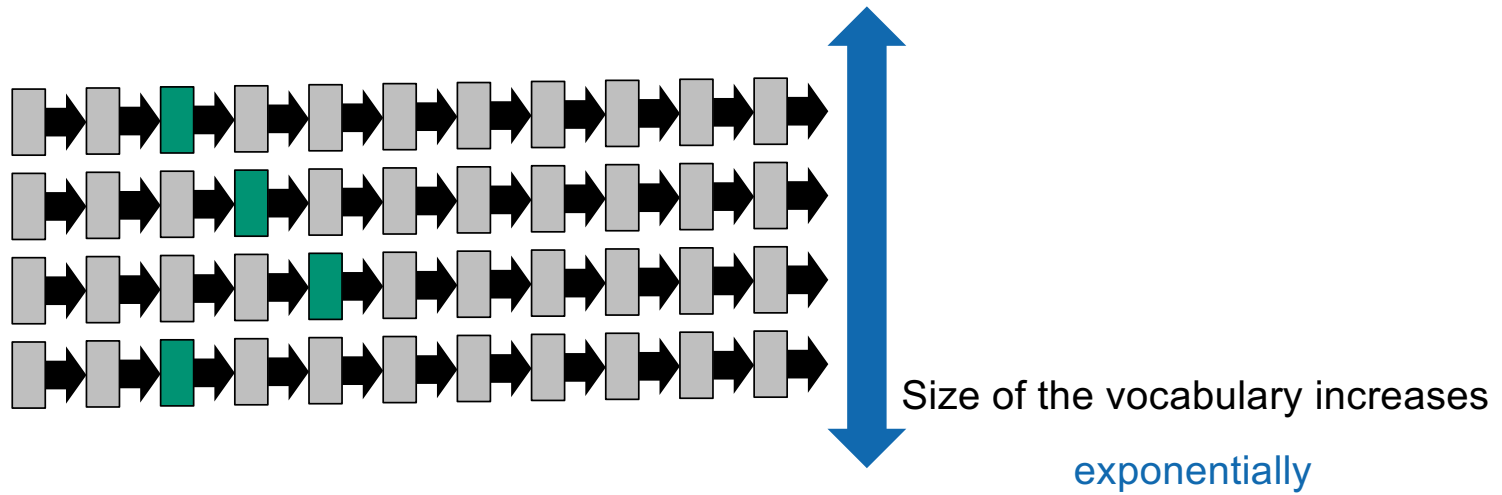


Inconvenient

Inconvenient



Inconvenient



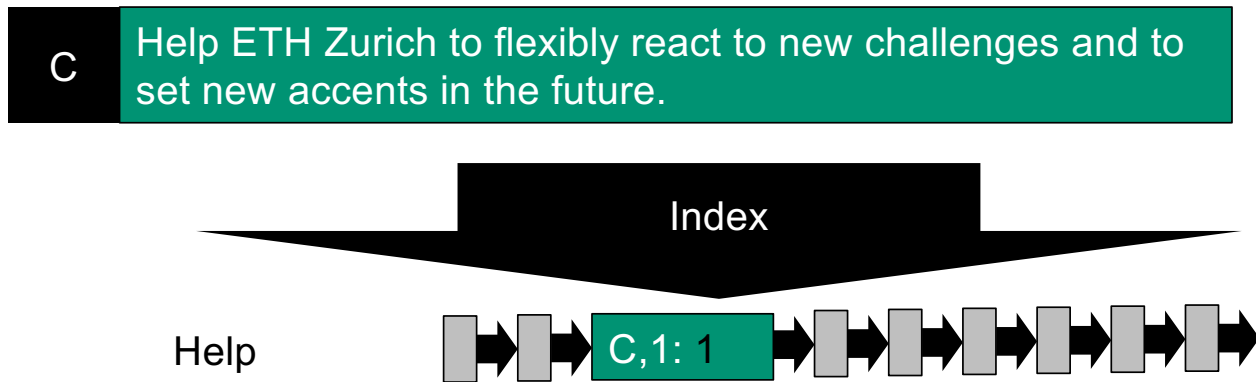
$$(\# \text{ Terms})^n$$

Positional index

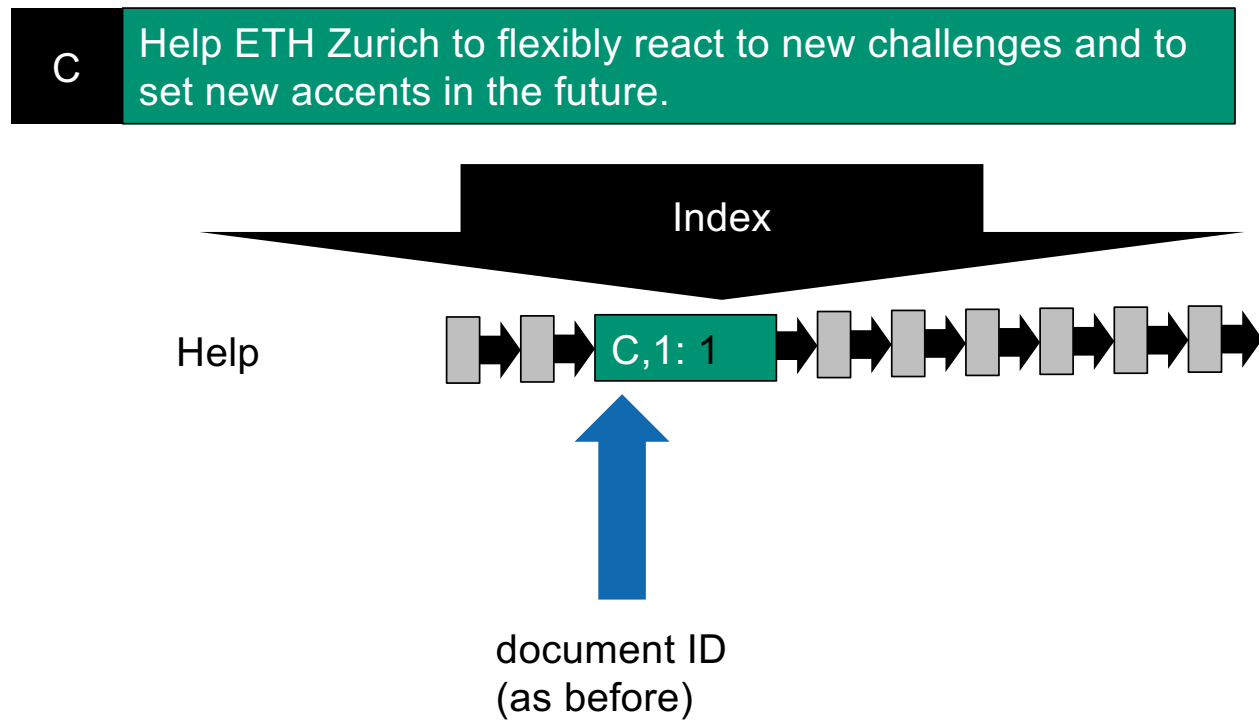
C

Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.

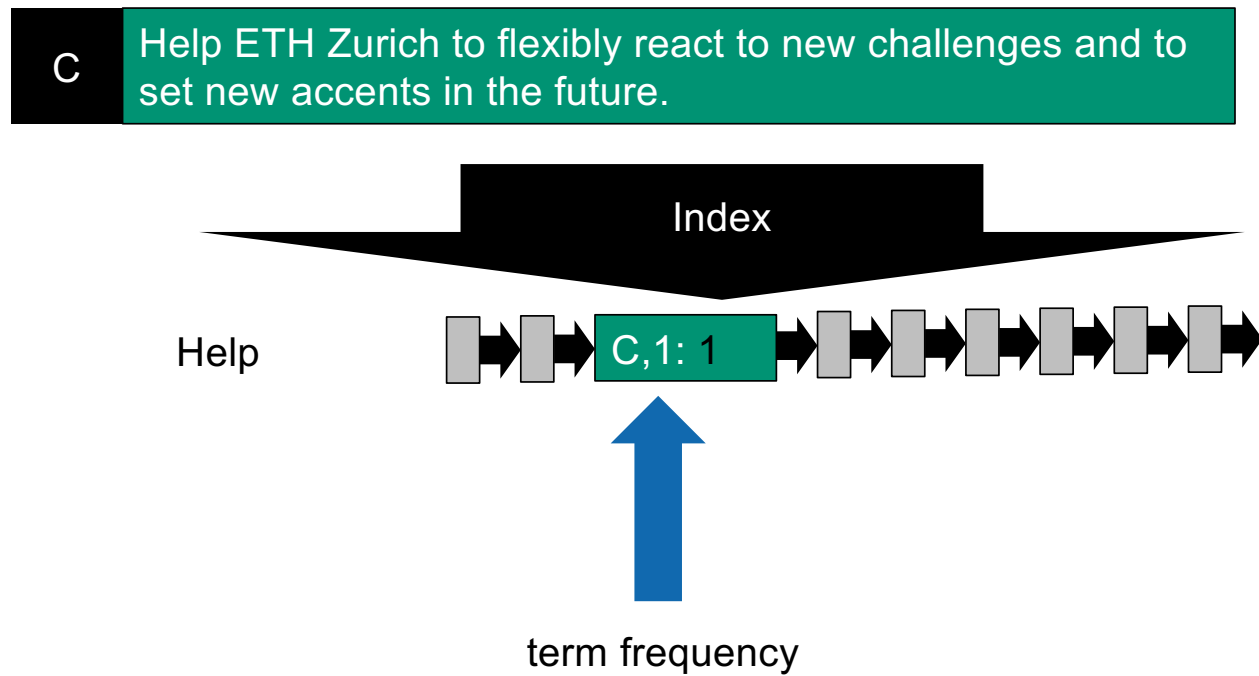
Positional index



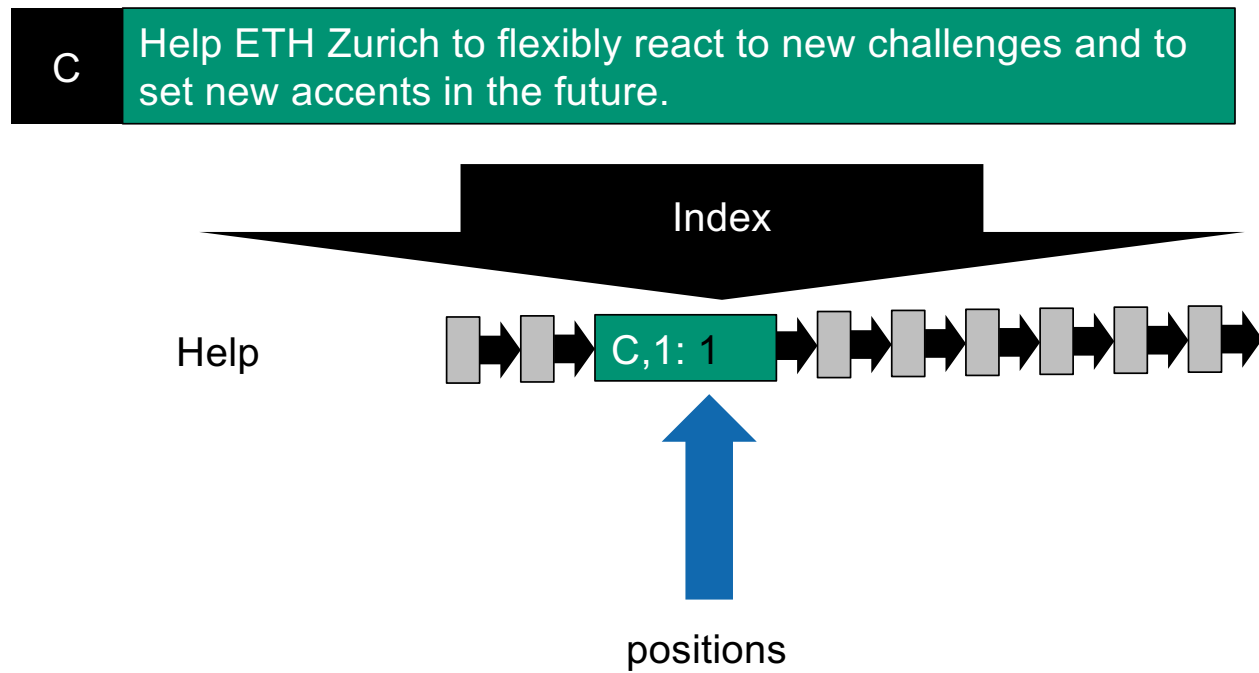
Positional index



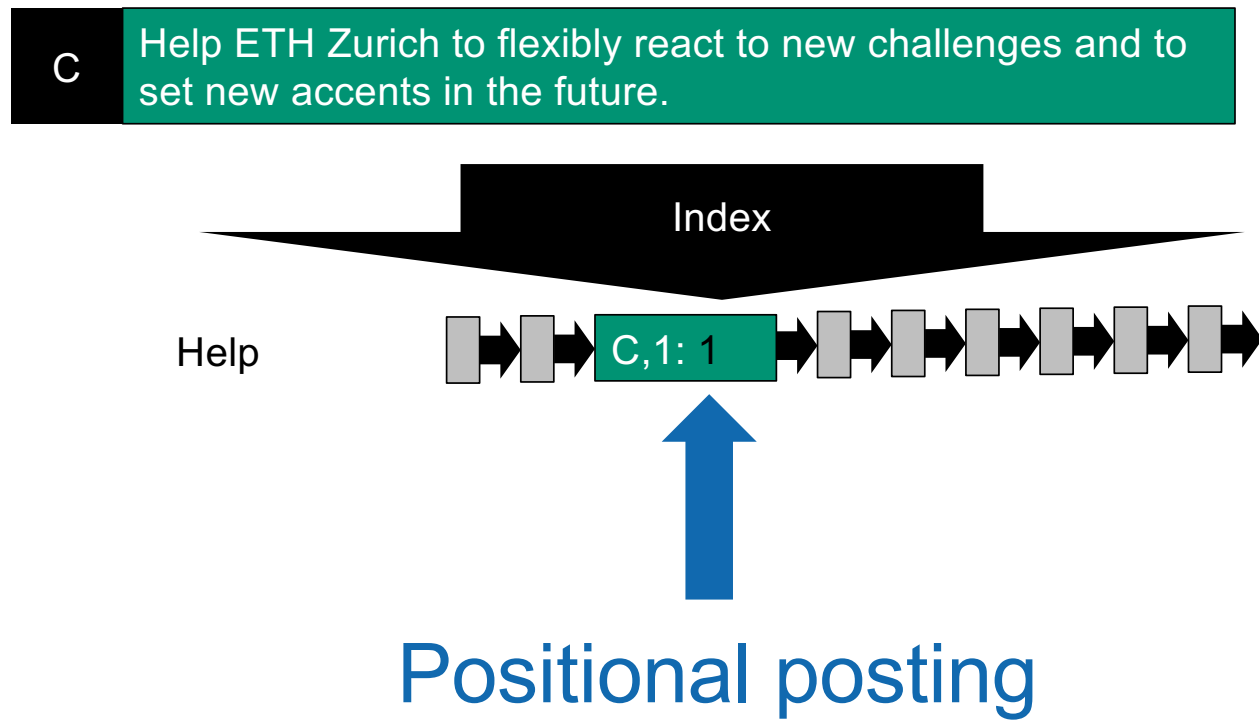
Positional index



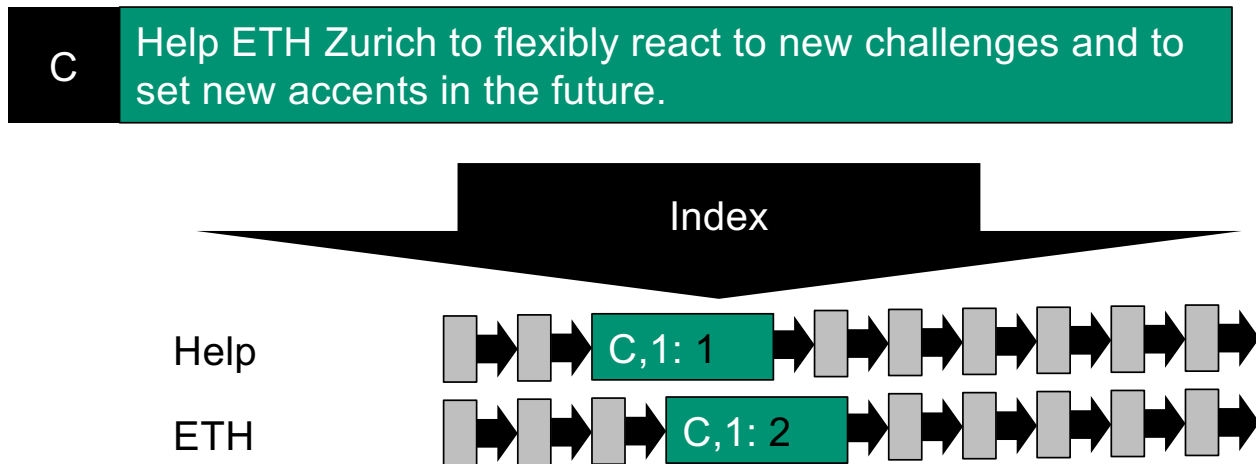
Positional index



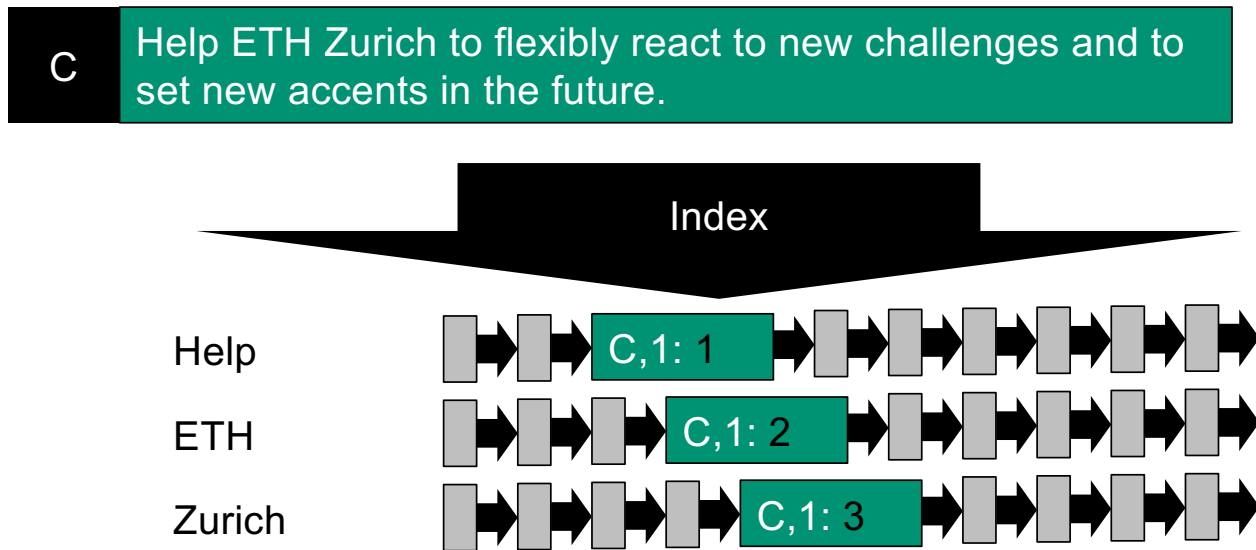
Positional index



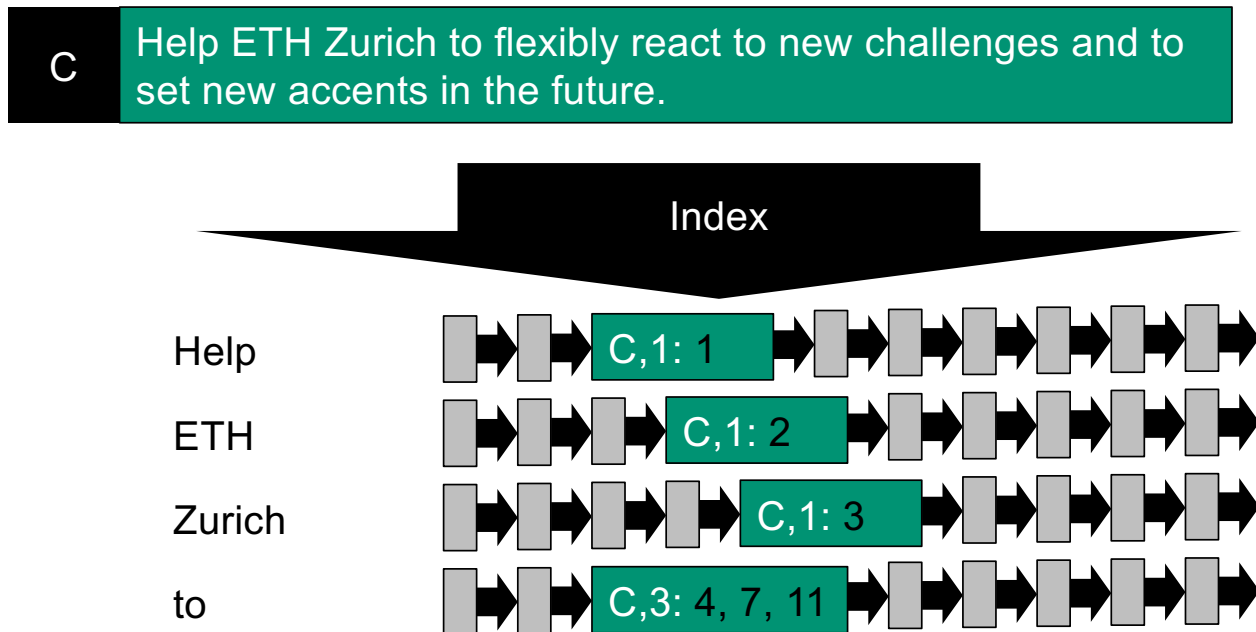
Positional index



Positional index

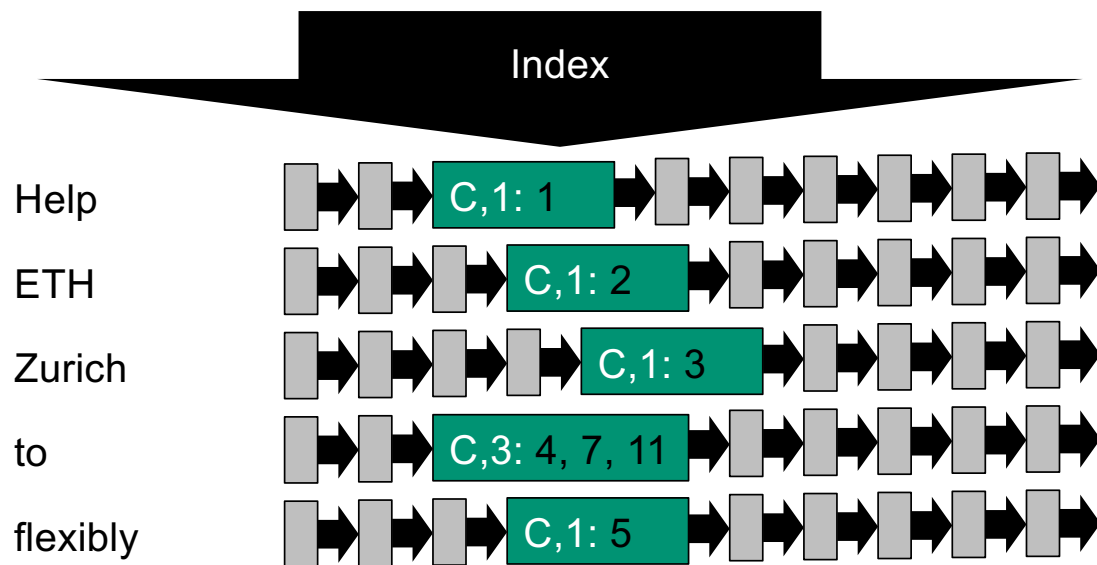


Positional index

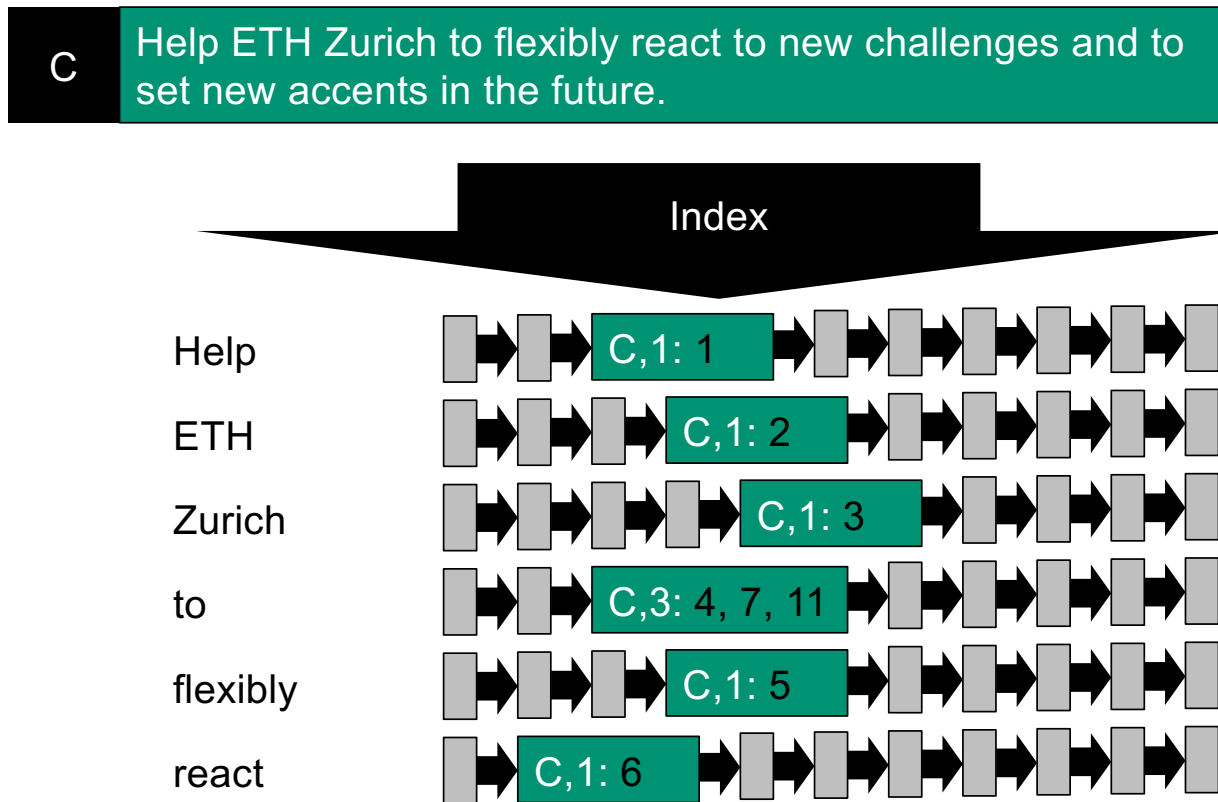


Positional index

C Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



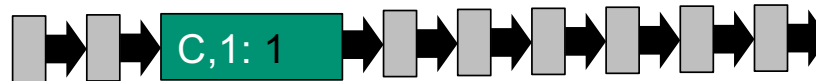
Positional index



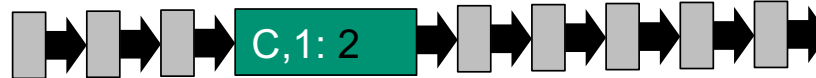
Positional index

"ETH Zurich"

Help



ETH



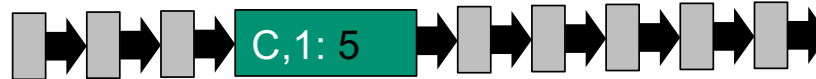
Zurich



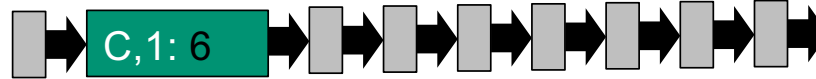
to



flexibly



react



Positional index

"ETH Zurich"

ETH



Zurich

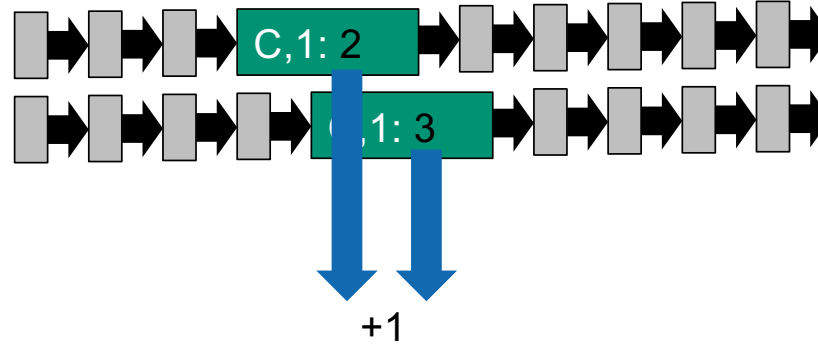


Positional index

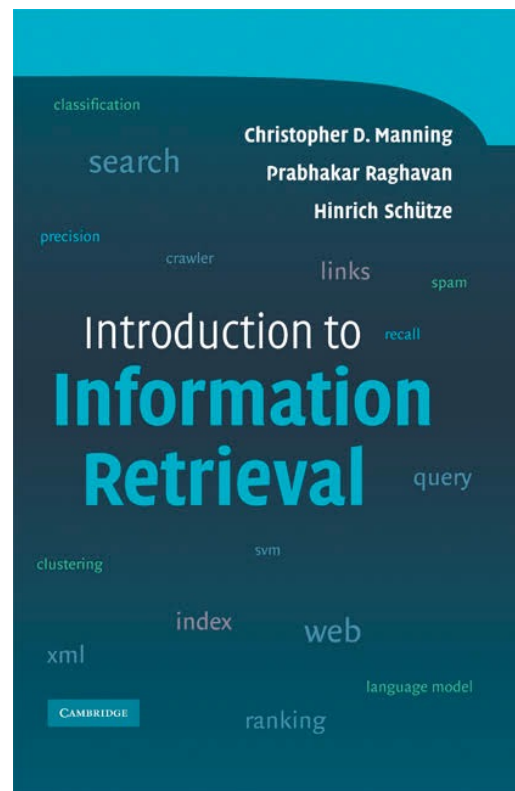
"ETH Zurich"

ETH

Zurich



This week's reading



Chapter 2

The Term Vocabulary and Postings Lists